Evaluating Joint Reliability for Observed Compositions

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Abstract

Conformal regions for compositions must distinguish joint coverage of an observed vector from coverage of class labels or latent probabilities. We compare split-conformal score geometries, training-only scaling and fixed-group calibration across 19 frozen synthetic cells with 100 independently regenerated/refitted trials each. A common original-simplex area and separate numerical audit make efficiency comparisons interpretable. With ten components, global log-ratio calibration has population coverage .9003 [95% interval .8975,.9030] but high-variance minority coverage .0511 [.0371,.0652]; fixed-group calibration raises minority coverage to .9096 [.9008,.9184]. At three components, the log-ratio-minus-Euclidean area difference changes from -.0439 in the interior to +.0498 near the boundary, with enlarged paired intervals excluding zero, while TV/Euclidean ordering remains unresolved. Group calibration tolerates the specified prevalence shift but returns the full domain in 31 of 100 scarce-minority calibrations. Nearly full-dimensional area also coexists with about 60% coverage on zero faces. These conditional findings support evaluating joint events, groups, applicability and numerical resolution together; they establish no universal score winner, latent-probability or real-data guarantee.

1 Introduction

A composition is a nonnegative vector summing to one. An uncertainty region around a compositional prediction is useful only if it covers the intended target. A classifier's probability vector can be used to predict a *class label*; an observed vector of proportions can instead be the response; a latent conditional probability vector is a third target. Covering one does not establish coverage of the others [Angelopoulos et al., 2020, Javanmardi et al., 2024].

We study joint coverage of a future *observed composition*, using split-conformal scores with separately fitted predictors. Exchangeable calibration and test scores give a finite-rank marginal guarantee [Papadopoulos et al., 2002, Lei et al., 2016, Angelopoulos and Bates, 2021]. This leaves practical choices of geometry, adaptation and calibration granularity. Equal marginal guarantees need not allocate uncertainty similarly across the simplex or across observable groups. Raw thresholds in different geometries cannot determine which region is smaller.

Our questions are: (Q1) how do geometry and training-only scaling affect joint reliability and original-simplex efficiency under exchangeability? (Q2) how much minority undercoverage can marginal validity hide, and what do scaling and group calibration cost? (Q3) how do boundary zeros, calibration scarcity and a specified calibration–test mismatch delimit applicability? Synthetic observations make the whole-vector event observable and let mechanisms vary separately from calibration. All methods share partitions and a fitted center, isolating score/calibration choices rather than comparing score-specific optimal predictors.

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Before confirmation, we hypothesized that joint scores attain their finite-rank marginal target while naive coordinate calibration need not; area orderings depend on the mechanism; and scaling or group calibration can reduce hidden minority failure at an efficiency or sample-size cost. Zeros and shift delimit domain and guarantee claims. We executed 19 frozen cells with 100 independently regenerated/refitted trials each, separately auditing numerical area uncertainty. The contribution is a reproducible evaluation with controlled positive, negative and unresolved findings, not a new conformal algorithm. The results show a geometry-ordering reversal, severe hidden minority failure, conservative full-domain outcomes with scarce group calibration, and the inability of full-dimensional area to diagnose zero-face coverage.

2 Related work and positioning

Calibration and adaptation. Inductive confidence machines separate proper training and calibration and already include learned residual scaling [Papadopoulos et al., 2002]. Split conformal prediction supplies economical marginal guarantees and locally weighted residual variants [Lei et al., 2016]. Coverage, set size, adaptation and calibration variability are separate evaluation questions [Angelopoulos and Bates, 2021]. We isolate these choices for simplex-valued responses; neither split calibration nor scaling is introduced here.

Regions for distributions versus label sets. The most direct permitted comparison is Javanmardi et al. [2024]: conformalized predictors from distribution-labeled data construct probability-simplex sets using distance and Dirichlet-likelihood scores, with discretized-simplex size evaluation and distinctions between noisy and clean labels. Thus generic simplex regions, distance scores and simplex-size evaluation are prior contributions. Our empirical scope is the interaction of geometry, calibration variability, observable-group reliability and boundary applicability under controlled mechanisms. APS/RAPS instead uses probability-vector features to predict class-label sets [Angelopoulos et al., 2020]; their cardinality measures a different response space.

Conditioning and calibration families. Class-conditional/clustered label calibration illustrates conditioning and calibration scarcity [Ding et al., 2023]; batch multivalid methods target stronger criteria over collections of groups and thresholds [Jung et al., 2022]. Our fixed-group audit establishes neither multivalidity nor pointwise conditional coverage. Jackknife+ changes the refitting/calibration construction and guarantee [Barber et al., 2019]; this split-score study does not compare all conformal algorithms. Statistical group coverage alone establishes neither fairness nor decision benefit [Cresswell et al., 2024].

Shift and composition geometry. Weighted conformal prediction addresses covariate shift under assumptions including invariant conditional response laws and suitable density ratios [Tibshirani et al., 2019]. Our unweighted prevalence-shift control diagnoses the limits of global exchangeable calibration, not that remedy. The ILR isometry on strictly positive compositions supplies the log-ratio geometry [Egozcue et al., 2003]; explicit smoothing changes the score when exact zeros occur.

3 Target and evaluation framework

3.1 Whole-vector coverage and finite-rank calibration

The observed response lies in

$$\Delta^{K-1} = \{y \in \mathbb{R}^K : y_j \geq 0,\ \textstyle\sum_{j=1}^K y_j = 1\}. \tag{1}$$

Separate training data determine a predictor $m(x)$ and a measurable scalar nonconformity score $s(x, y)$, without fitting or tuning on calibration responses. For calibration scores $S_1, \ldots, S_n$ and error level $\alpha \in (0, 1)$, define

$$k = \lceil (n+1)(1-\alpha) \rceil, \qquad q = \begin{cases} S_{(k)}, & k \leq n, \\ +\infty, & k = n+1, \end{cases} \qquad C_\alpha(x) = \{y \in \Delta^{K-1} : s(x, y) \leq q\}. \tag{2}$$

Under exchangeable calibration/new scores, this gives marginal whole-vector coverage at least $1 - \alpha$ [Angelopoulos and Bates, 2021]. Ties use $\leq$; an infinite threshold denotes the full domain.

Conditional on independent training, the guarantee averages over calibration and a future observation; it does not hold for every realized calibration set or feature value. Appendix A derives the bound and the narrower IID continuous-score calibration-variability reference.

Coordinatewise events are separate. Calibrating each coordinate at error α supplies no joint $1-\alpha$ guarantee; allocating α/K gives a Bonferroni joint guarantee by the union bound, without coordinate independence. Intersecting an ambient region with the simplex preserves coverage because the target lies there.

3.2 Reliability, efficiency, and uncertainty

For independent fitted/calibrated trial r with M_r test observations, let $I_{ri} = \mathbb{1}\{Y_{ri} \in C_{\alpha,r}(X_{ri})\}$. Report

$$\widehat{c}_r = M_r^{-1} \sum_{i=1}^{M_r} I_{ri}, \qquad \overline{c} = R^{-1} \sum_{r=1}^{R} \widehat{c}_r, \tag{3}$$

and the corresponding group estimates with their actual denominators. Mean precision and across-trial spread describe different properties: a precise average does not imply stability of one calibration. Tests sharing a fit and threshold are not independent replicates for an unconditional pooled binomial interval [Angelopoulos and Bates, 2021]. Methods share partitions and use paired trial differences; Section 4 specifies the outer-trial and numerical uncertainty procedures.

Efficiency uses a common original-space measure,

$$a_r(x) = \frac{\mathrm{vol}_{K-1}(C_{\alpha,r}(x))}{\mathrm{vol}_{K-1}(\Delta^{K-1})}, \qquad A = \mathbb{E}_r \mathbb{E}_{X \sim P_{\mathrm{test}}}[a_r(X)]. \tag{4}$$

Ambient K-dimensional volume is zero for all simplex regions; raw radii in different geometries are also incomparable. Compare common area alongside achieved joint coverage: an under-covering smaller region does not establish better calibrated efficiency. Location subsets must represent the declared test-X distribution and use test-group weights, especially under shift. Location sampling and integration error remain distinct uncertainty sources. Zero integration hits do not establish zero area, and unresolved numerical distinctions do not support rankings. Full-dimensional area assigns zero measure to faces, so it cannot alone describe informativeness for face-supported or atomic responses.

4 Implemented comparisons and controlled mechanisms

Score families and controls. All methods share a positive fitted center $m(x)$. With $B \in \mathbb{R}^{K\times(K-1)}$ satisfying $B^T B = I$ and $B^T \mathbf{1} = 0$, define $z(y) = B^T \log y$ on the positive simplex [Egozcue et al., 2003]. The implemented scalar scores are

$$s_{\mathrm{TV}}(x,y) = \tfrac{1}{2}\|y - m(x)\|_1, \qquad s_{\mathrm{L2}}(x,y) = \|y - m(x)\|_2, \tag{5}$$

$$s_{\mathrm{H}}(x,y) = \tfrac{1}{\sqrt{2}}\|\sqrt{y} - \sqrt{m(x)}\|_2, \qquad s_{\mathrm{ILR}}(x,y) = \|z(y) - z(m(x))\|_2. \tag{6}$$

Each score uses Equation (2). Coordinate controls instead calibrate $|Y_j - m_j(X)|$ separately and intersect all resulting intervals with Δ^{K-1}, either at error α (naive) or α/K (Bonferroni). Naive calibration controls individual coordinates, not the desired joint event. Heterogeneous cells additionally use $s_{\mathrm{ILR}}(x,y)/\widehat{\sigma}_{G(x)}$ with training-only scales, or separate ILR thresholds for each fixed group $G \in \{0,1\}$ (Mondrian). Thus comparisons isolate score geometry, scaling, and calibration granularity rather than different fitted centers.

For actual-zero responses, unsmoothed ILR is recorded as inapplicable to the entire mechanism. The operative ILR-smoothed score replaces both arguments by

$$T_\epsilon(y) = \frac{y + \epsilon\mathbf{1}}{1 + K\epsilon}, \qquad \epsilon > 0. \tag{7}$$

The region consists of original candidates y whose transformed score is below threshold. It includes the original zero-valued responses in its coverage denominator. The primary $\epsilon = .001$ and sensitivities $10^{-6}, .01$ define different scores; their location fit is held fixed. Finite-threshold full-domain status and infinite-threshold frequency are distinct diagnostics. Where full-domain status for smoothed ILR cannot be certified, it is recorded as unknown rather than inferred from an estimated area near one.

Predictor and data mechanisms. The covariates are $X = (u, v, G)$, with independent $u, v \sim \mathrm{Uniform}[-1, 1]$ and $G \sim \mathrm{Bernoulli}(.1)$. A deterministic feature/coefficient rule generates a positive base center $\mu(X)$ (Appendix B). Five mechanisms contrast: (i) isotropic logistic-normal noise of scale $.45$ in ILR coordinates; (ii) group-heterogeneous logistic-normal scales $.25$ and 1.25; (iii) component-0 logit shifted by 3 with logistic-normal scale $.8$; (iv) a Dirichlet distribution of total concentration 15 around a floored version of the shifted center; and (v) logistic-normal scale $.8$, followed independently with probability $.25$ by setting one uniformly selected component exactly to zero and renormalizing. These are specified generative mechanisms, not estimated characteristics of a real dataset.

A trial draws 2,000 proper-training observations, 499 calibration observations and 2,000 tests. The first 1,500 training observations fit ridge regression of $z(Y)$ on $(1, u, v, \sin(\pi u), uv, G)$ with penalty $.001$ and an unpenalized intercept. In the zero mechanism only, the regression target uses $T_{.001}(Y)$. Inverse ILR gives the shared positive center $m(x)$. The remaining 500 training observations fit groupwise RMS ILR residual scales; an absent group uses overall RMS, with floor 10^{-8}. No calibration or test response fits a location or scale. This center estimates a log-ratio location, generally different from the Euclidean conditional mean. Geometry conclusions are conditional on this shared predictor rather than per-score optimally chosen models.

Prespecified study matrix and contrasts. The pre-outcome frozen manifest specifies 19 cells and 100 independently regenerated/refitted trials per cell. Ten core cells cross five mechanisms with $K = 3, 10$. At $K = 3$, interior and heterogeneous mechanisms additionally use calibration sizes 99 and $1{,}999$ and nominal error $.05$. Two zero-mechanism cells vary ϵ, and one heterogeneous stress cell changes test-group prevalence from $.1$ to $.4$. This stress preserves the complete within-group law: it is a specific covariate shift, not a change in $Y \mid X$. Fixed-group calibration retains its within-group and mixture guarantee for this change; unweighted global calibration has no such general protection (Appendix A).

The declared contrasts compare TV/L2/Hellinger with ILR, naive coordinates with Bonferroni, and global ILR with scaled and Mondrian ILR in heterogeneous cells. Sensitivities compare with their matched base cell. All 19 cells and their prescribed numerical audits completed; pilots are excluded from confirmation. The scientific seeds are offset by $10{,}000{,}000$ from pilot seeds; all methods within a trial share partitions, and matched sensitivity arms share the declared seed. Exact cell seeds, score applicability, and execution provenance are machine-readable under `analysis/`.

Efficiency estimation and interpretation policy. Common area is measured only at $K = 3$; $K = 10$ supports coverage and applicability without a high-dimensional volume ranking. For each fitted trial, area is evaluated at the first 32 test locations in each observed group. Test order is independent of outcomes, so these form groupwise random subsets. Their means are weighted by actual test-group proportions. An absent test group contributes zero weight; a missing area in a positive-weight group remains undefined. Transformed-region integration uses 2^{14} scrambled Sobol points with the common-simplex Jacobian (Appendix C). Trial/group/location keys separate integration randomness across trials and preserve numerical pairing across methods. Integration maps still differ across geometries.

Outer-trial summaries use means, trial SDs and two-sided Student-t 95% intervals, with 5,000-draw percentile trial-bootstrap sensitivity intervals (seed 90210). Paired contrasts use within-trial differences. These intervals include changing fits, calibration samples, tests and area-location subsets; they do not bound shared integration bias. A separate, prespecified audit recomputes matched locations at 2^{16} points with four scrambles. An empirical numerical allowance enlarges paired statistical intervals (Appendix C); it is not a uniform numerical confidence bound. Missing audits or enlarged intervals crossing zero leave an area ranking unresolved. Pre-outcome practical margins are $.01$ in coverage and $.005$ in normalized area; smaller effects do not establish a useful ranking. Minority-coverage mean half-widths above $.02$ are flagged as imprecise. These are study interpretation choices, not field-wide standards or equivalence tests.

5 Results

All 19 frozen cells completed 100 regenerated/refitted trials each, with 2,000 tests per trial: 44,600 method/group records, including 2,000 explicit unsmoothed-ILR inapplicability records. No pilot

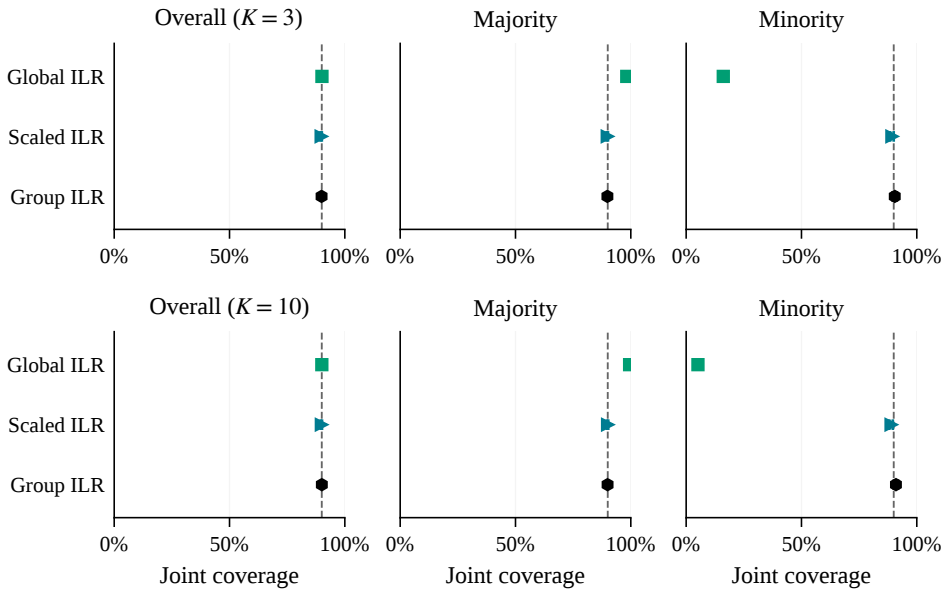


Figure 1: Overall coverage can conceal severe minority undercoverage. Measured whole-vector event $Y \in C_{0.1}(X)$ in the heterogeneous mechanism, with observable majority/minority prevalence 0.9/0.1 and a common inverse-ILR fitted center. Global ILR, training-fitted scaled ILR, and fixed-group ILR use the same samples within each trial. Points and bars show mean and pointwise 95% Student-t confidence intervals over 100 independent regenerated/refitted trials per dimension (2000 training, 499 calibration and 2000 test observations each); bars quantify uncertainty of the mean, not trial-to-trial variation, and may be narrower than the symbols. Minority count ranges across trials: $K = 3$: test 166–231, calibration 33–67; $K = 10$: test 163–237, calibration 36–73. Dashed lines mark nominal 90%; every panel uses the same 0–100% scale. Scaling improves the measured minority allocation but does not grant exact group validity. These synthetic core results do not establish real-data or arbitrary-shift performance; area comparisons use the separately audited enlarged paired intervals.

enters confirmation. Brackets below are pointwise Student-t 95% intervals over independent trials, with pairing for contrasts; they are not simultaneous theorem tests or guarantees for each calibration sample. Appendix I retains every declared method/group entry, domain outcome and denominator; Appendix G shows all 32 prespecified measured cases.

Q1: coordinate calibration and joint reliability. Naive coordinate calibration gives interior joint coverage .7806 [.7766, .7847] at $K = 3$ and .3215 [.3176, .3254] at $K = 10$ (Table 1). In the latter cell the ten coordinate means range from .8974 to .9009: near-nominal components coexist with severe joint failure. Bonferroni instead gives .9215 [.9188, .9242] and .8981 [.8950, .9013]. The control fails the desired event, not its coordinatewise interpretation. Operative joint scores are near nominal in the core cells, but not every pointwise interval contains .9: interior $K = 10$ L2 is .8967 [.8936, .8999]. This small shortfall is below the .01 practical margin; many finite descriptive intervals do not refute the exchangeable rank theorem.

Q2: marginal coverage hides minority failure. Figure 1 contrasts global, fitted-scale and fixed-group ILR. At $K = 3$, global population coverage .9008 [.8980, .9036] masks minority coverage .1610 [.1534, .1686]. At $K = 10$, population .9003 [.8975, .9030] masks minority .0511 [.0371, .0652]. Scaling raises minority coverage to .8921 [.8842, .8999] and .8885 [.8808, .8961]; group calibration gives .9044 [.8955, .9134] and .9096 [.9008, .9184]. Group-minus-global paired minority improvements are .7435 [.7317, .7553] and .8584 [.8429, .8740], exceeding the declared margin with mean half-widths below .02. Scaling helps here but its minority means remain below .9; only fixed-group calibration has the stated group theorem.

Table 1: Whole-vector coverage under coordinate calibration (percent). Means [pointwise 95% Student-t intervals] over 100 independently regenerated/refitted trials per cell; $n_{\text{train}} = 2000$, $n_{\text{cal}} = 499$, $n_{\text{test}} = 2000$, joint target 90%. Marginal coordinates calibrate each coordinate at 90%; Bonferroni uses $1 - 0.1/K$ per coordinate. All methods in a cell share fitted centers and samples. The intervals quantify mean coverage, not trial-to-trial spread or simultaneous inference. Coordinate-specific means and intervals appear in the companion source CSV and coverage figure.

Mechanism	K	Marginal coordinates	Bonferroni
Interior	3	78.06 [77.66, 78.47]	92.15 [91.88, 92.42]
Interior	10	32.15 [31.76, 32.54]	89.81 [89.50, 90.13]
Heterogeneous	3	81.58 [81.11, 82.05]	94.06 [93.84, 94.28]
Heterogeneous	10	55.20 [54.43, 55.97]	92.23 [92.07, 92.38]
Near boundary	3	81.80 [81.46, 82.15]	93.63 [93.40, 93.85]
Near boundary	10	46.56 [46.20, 46.93]	91.21 [90.93, 91.49]
Dirichlet	3	79.58 [79.20, 79.97]	92.78 [92.53, 93.02]
Dirichlet	10	34.18 [33.76, 34.60]	90.76 [90.45, 91.07]
Exact zeros	3	78.43 [78.05, 78.82]	92.56 [92.33, 92.79]
Exact zeros	10	26.69 [26.34, 27.04]	89.90 [89.60, 90.20]

Table 2: Paired original-simplex area differences at $K = 3$ (percentage points of the whole simplex). Each entry is mean [95% paired Student-t interval enlarged on both sides by the separate measured numerical allowance], over 100 independent trials. ILR is explicitly smoothed with $\epsilon = .001$ in the exact-zero arm. R: resolved sign with absolute mean difference at least 0.5 area percentage points; U: enlarged interval contains zero or no audit; S: resolved sign below that practical margin. These are geometric flags, not superiority at a common group guarantee. Negative means smaller area for the first score; achieved coverage must be considered separately. TV/Euclidean remains unrankable in every core mechanism. Allowances are finite sensitivity diagnostics, not uniform bounds; zero faces have zero full-dimensional area.

Mechanism	ILR minus Euclidean	TV minus Euclidean
Interior	-4.39 [-4.76, -4.02] R	0.07 [-0.15, 0.30] U
Heterogeneous	-2.49 [-2.91, -2.07] R	-0.13 [-0.34, 0.08] U
Near boundary	4.98 [4.55, 5.40] R	-0.09 [-0.23, 0.05] U
Dirichlet	17.71 [17.02, 18.39] R	-0.02 [-0.16, 0.13] U
Exact zeros	23.99 [23.04, 24.94] R	0.29 [-0.21, 0.79] U

Q1–Q2: geometry ordering and adaptation costs. Figure 2 places achieved joint coverage beside common area. Table 2 uses paired statistical intervals enlarged by the separate numerical allowance. ILR-minus-L2 mean area is $-.04394$ $[-.04764, -.04024]$ in the interior, but $+.04977$ $[.04554, .05401]$ near the boundary and $+.17705$ $[.17023, .18388]$ under the Dirichlet generator. These resolved reversals rule out a universal score ordering for this shared predictor. Every core-K3 TV/L2 comparison remains unresolved; interior/heterogeneous ILR versus Hellinger is also unresolved. A smaller naive-coordinate region with joint undercoverage is not better calibrated efficiency.

In the heterogeneous $K = 3$ cell, fitted-scale and group calibration increase population area over global ILR by .01222 [.00608, .01835] and .01384 [.00767, .02002], respectively (enlarged intervals). This measured cost accompanies substantially better minority allocation; it does not compare identical group guarantees. The scaled-versus-group area contrast is unresolved. The full audit covers 3,008 method/center regions and 3,616 paired numerical contrasts; the largest empirical allowance is .004229. Among 339 reported area contrasts, 76 remain unresolved and 258 meet the geometric practical-resolution flag. Those flags alone do not establish useful efficiency at the relevant coverage; the remaining sign-resolved small effects do not meet the practical margin. No K10 area is measured.

Q3: the specified shift and calibration scarcity. Changing only minority test prevalence from .1 to .4 reduces global ILR population coverage to .6539 [.6499, .6580]. Fixed-group ILR gives .9020 [.8982, .9059]; scaling gives .8974 [.8938, .9010]. The complete within-group law stays fixed,

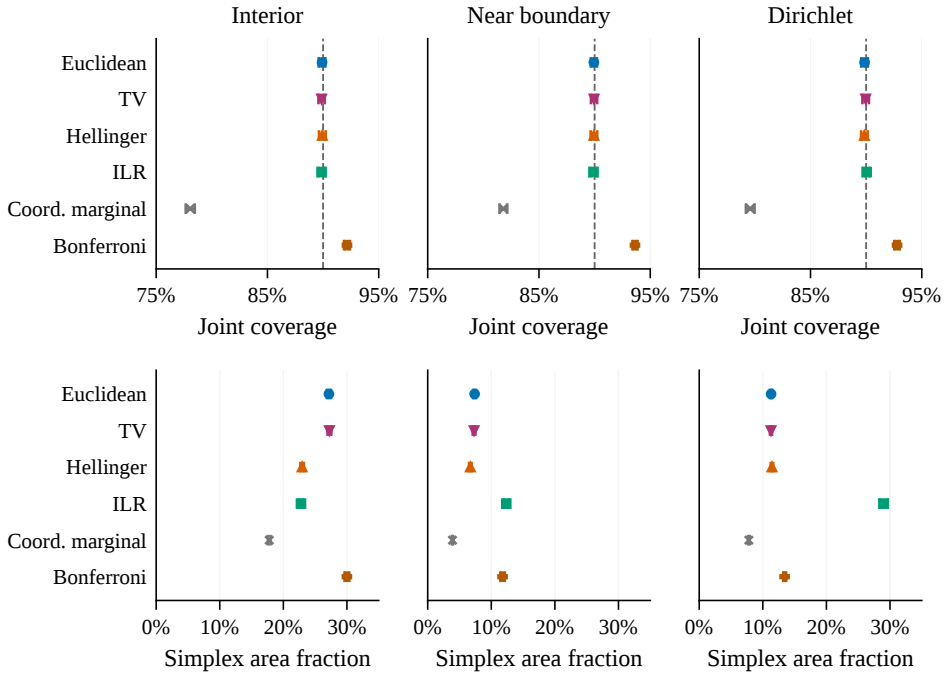


Figure 2: Coverage and common area depend on score geometry and mechanism. All panels use the original observed-composition event at $K = 3$, nominal 90%, with the same fitted inverse-ILR centers and partitions across methods. Points/bars are means and pointwise 95% Student-t intervals over 100 independent regenerated/refitted trials per cell (train/calibration/test 2000/499/2000). Coverage axes focus on 75–95%; area axes on 0–35% of the full simplex. Mean normalized area averages the first 32 randomly ordered test locations per observable group (64 per trial here), weighted by observed test-group prevalence. These area bars describe outer-trial variation; ranking additionally requires the enlarged paired intervals in the companion contrast table, not overlap of these bars. The full simplex has area fraction one. A smaller coordinate-marginal region accompanies undercoverage. No universal score winner or $K = 10$ area claim follows.

so this agrees with the group-mixture derivation, not arbitrary-shift robustness (Figure 12). With total calibration size 99, group ILR returns an infinite threshold/full domain for the minority in 31/100 trials; minority coverage is .9493 [.9364, .9622]. This conservatism is retained, not discarded as a failure. At size 1999, its minority mean is .8943 [.8882, .9003], and population mean .8978 [.8959, .8998]: more data does not force every finite-run mean above nominal.

For interior ILR, across-trial coverage SD decreases from .03086 at calibration size 99 to .01562 at 499 and .01065 at 1999, with finite-test noise included. The size 99 mean .8939 [.8877, .9000] is another small pointwise shortfall. The $\alpha = .05$ interior arm gives .9497 [.9475, .9520], targeting .95. Tables 7–8 retain calibration-size/nominal effects and distinguish SD from mean uncertainty. Some scarce-group comparisons have minority mean half-width above .02; neither wide nor overlapping intervals establish equivalence.

Q3: zeros, smoothing and nearly full area. Unsmoothed ILR remains inapplicable to the entire actual-zero arm. The operative $\epsilon = .001$ score covers original compositions at .8995 [.8968, .9022] for $K = 3$ and .8995 [.8964, .9025] for $K = 10$. Yet $K = 3$ zero-face coverage is .5985 [.5878, .6092], positive-response coverage is 1 in every trial, and mean full-dimensional area is .997125 [.996953, .997297]. L2 zero-face coverage is .6926 [.6854, .6998] at population coverage .9017 [.8988, .9046]. Response-defined slices are descriptive, not test-observable groups with a conditional theorem.

Across $\epsilon = 10^{-6}, .001, .01$, smoothed-ILR face coverage stays near .60 (Table 9). At 10^{-6}, mean area is 1.000018 $[.999994, 1.000041]$, an estimate saturated near one. The audit retains 159 small out-of-range trial/group mean estimates (maximum 1.000374) unclipped. These are numerical warnings, not true areas above one, proof of full-domain inclusion, or resolved cross-epsilon rankings. Faces have zero full-dimensional measure: near-full area cannot characterize their uncertainty. Full-domain bounds, unknown-status rates and infinite thresholds are reported separately.

6 Discussion and limitations

The study answers the reliability question through four distinctions: the joint target event, the population over which coverage averages, the score's domain and a common geometric measure with explicit numerical resolution. The controlled area reversal supports mechanism-dependent choice rather than a universal geometry. In these experiments ILR is more area-efficient than L2 for the interior mechanism; the ordering reverses near the boundary and under the specified Dirichlet generator. TV/L2 differences remain unresolved under the declared rule. These comparisons are conditional on the common fitted log-ratio center, not score-specific optimal predictors.

For a marginal target under exchangeability, a globally calibrated joint score supplies the stated guarantee. When a scientifically relevant observable group has different variability, our core results favor auditing it explicitly; aggregate coverage can be misleading. Training-only scaling substantially improves this minority but does not grant exact group validity. Fixed-group calibration is appropriate when within-group exchangeability is credible and calibration support is adequate; the scarce-group full-domain outcomes quantify its possible information cost. Its prevalence-shift protection requires invariant within-group laws, not arbitrary distribution changes.

Original-zero membership must be defined rather than silently dropping responses. Smoothing produces a valid scalar score on original candidates but changes geometry, and full-dimensional area cannot describe uncertainty along response faces. These slices depend on the response and have no stated conditional guarantee. Finite integration remains imperfect:159 small range warnings are retained, empirical allowances are nonuniform,76 area contrasts remain unresolved, and no K10 area ordering is attempted.

The target is an observed composition, not its conditional mean, a class label or latent probabilities. All data are synthetic; five mechanisms, two dimensions and one shared ridge predictor do not establish deployed-classifier, topic-model or real-data performance. The study compares specified split scores and fixed groups, not all conformal families or arbitrary subgroups. Pointwise intervals across correlated comparisons are descriptive rather than familywise tests; minor nominal shortfalls remain visible. Group reliability is a statistical diagnostic, not proof of fairness or downstream benefit. Controlled synthetic evidence makes these evaluation failures checkable, while application-specific assessment remains necessary.

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A Rank coverage, calibration variability, and the specified shift

Condition on proper-training data and all independently chosen training randomness. Assume the resulting scalar calibration/new scores $S_1, \ldots, S_n, S_{n+1}$ are exchangeable, and use k and q from Equation (2). Break ties by independent random ordering for this argument only. The new score has uniform rank among $n+1$ scores. If its rank is at most $k \le n$, then $S_{n+1} \le S_{(k)}$ among the calibration scores. Therefore

$$\Pr\{Y_{n+1} \in C_\alpha(X_{n+1}) \mid \text{training}\} \ge \frac{k}{n+1} \ge 1-\alpha. \tag{8}$$

For distinct scores the first inequality is equality. Including ties with $\le$ can only increase coverage. If $k = n+1$, the infinite threshold gives coverage one. Intersecting with the simplex preserves this event because the target lies there. A fixed positive training-derived scale and deterministic transformation in Equation (7) preserve score exchangeability. The proof averages over calibration and a new observation; it does not condition on a realized calibration sample or establish pointwise conditional coverage.

For each coordinate j, calibration at α/K gives marginal coordinate miscoverage at most α/K. The union bound then bounds any-coordinate miscoverage by α, with no coordinate-independence assumption. Calibrating each at α instead gives only the bound $K\alpha$; it supplies no joint $1-\alpha$ guarantee.

For fixed observable group g, condition also on its realized calibration count n_g. Exchangeability of within-group scores yields the same bound for a new group-g observation using its corrected rank, including infinity for insufficient calibration ranks. In the specified stress arm, the law of $(X, Y) \mid G = g$ is unchanged; only the test weights $w_g = \Pr_{\text{test}}(G = g)$ change. Thus the group bounds imply

$$\Pr_{\text{test}}(Y \in C(X) \mid \text{training}, \{n_g\}) = \sum_g w_g \Pr_{\text{test}}(Y \in C(X) \mid G = g, \text{training}, \{n_g\}) \ge 1-\alpha. \tag{9}$$

The probability still averages over within-group calibration values and the future observation. This protection for fixed-group calibration does not extend to an arbitrary change inside groups. The stress arm does not evaluate density-ratio-weighted conformal prediction or general conditional-response shift.

Continuous-score diagnostic. Under the stronger assumption of conditionally IID scores with continuous CDF F, the transformed scores are IID uniform. For $k \le n$, true coverage conditional on the realized threshold is $P = F(S_{(k)})$, hence

$$P \sim \text{Beta}(k, n+1-k), \quad \mathbb{E}[P] = \frac{k}{n+1}, \quad \text{Var}(P) = \frac{k(n+1-k)}{(n+1)^2(n+2)}. \tag{10}$$

This is the uniform order-statistic distribution, obtained by counting whether at least k of n uniform scores lie below a given value. For M conditionally independent tests, the covered count conditional on P is binomial. Total variance gives

$$\text{Var}(\widehat{c}) = \text{Var}(P) + \frac{\mathbb{E}[P(1-P)]}{M}. \tag{11}$$

The saved uniform-score simulation and Figure 3 check these references. IID continuity is required for this Beta law; exchangeability alone suffices for Equation (8) but not this distributional identity.

B Exact generator and fitting specification

Let $j = 0, \ldots, K-1$, $\phi(X) = (1, u, v, \sin(\pi u), uv, G)$, and let the component coefficient vector be

$$a_j = \left(.4\cos j,\ .7\cos\frac{2\pi j}{K},\ .5\sin\frac{2\pi j}{K},\ .3\cos\frac{4\pi j}{K},\ .2\sin\frac{4\pi j}{K},\ .2\cos\frac{2\pi j}{K}\right)^T. \tag{12}$$

Write $\eta_j = \phi a_j$ and $\mu = \text{softmax}(\eta)$. Interior, heterogeneous and zero mechanisms use this center. Boundary and Dirichlet mechanisms first replace η_0 by $\eta_0 + 3$. For logistic-normal mechanisms,

$$Y = \text{softmax}\{B[z(\mu) + \sigma_G Z]\}, \qquad Z \sim N(0, I_{K-1}), \tag{13}$$

with the scales in Section 4. All observations use fresh independent generator draws. For the Dirichlet mechanism set $\mu_D = .8\mu + .2\mathbf{1}/K$, sample independent $W_j \sim \mathrm{Gamma}(15\mu_{Dj}, 1)$, and take $Y_j = W_j / \sum_l W_l$. For the zero mechanism, independently with probability .25 select J uniformly from $\{0, \ldots, K-1\}$, set the logistic-normal vector's Jth entry to zero and divide by its remaining sum. The zero/positive slices condition on the realized response; they are descriptive audits rather than test-observable calibration groups.

Let Φ contain the first 1,500 training feature rows and Z_{train} their ILR response rows, with fixed zero smoothing where stated. The location coefficient matrix and inverse transform are

$$\begin{aligned} \widehat{A} &= (\Phi^T\Phi + \Lambda)^{-1}\Phi^T Z_{\mathrm{train}}, \quad \Lambda = \mathrm{diag}(0, .001, .001, .001, .001, .001), \\ m(x) &= \mathrm{softmax}\{B\widehat{A}^T\phi(x)^T\}. \end{aligned} \tag{14}$$

On the remaining 500 training observations, take the RMS of ILR residual norms within each group, using overall RMS for an absent group and flooring at 10^{-8}. These observations do not fit the location. The predictor is intentionally shared across geometries and is misspecified for some mechanisms; conformal validity does not require correct specification. Generator/training implementations are `analysis/src/compbench/data.py`; the orthonormal basis and transforms are in `geometry.py`. The frozen manifests retain all trial seeds, split sizes and sensitivity choices.

C Common-simplex integration and empirical numerical allowance

Let $d = K-1$, $V_d = \pi^{d/2}/\Gamma(d/2+1)$, and $\mathrm{vol}_d(\Delta^{K-1}) = \sqrt{K}/d!$. These formulas use Hausdorff measure in the affine hyperplane. The primary evaluation integrates only $K = 3$ cases.

Euclidean and coordinate regions. A tangent-space Euclidean ball of radius r has volume $V_d r^d$. Its normalized in-simplex area is

$$a_{\mathrm{L2}} = \frac{d!}{\sqrt{K}} V_d r^d \, \mathbb{E}_U[\mathbb{1}\{m + rBU \geq 0\}], \qquad U \sim \mathrm{Uniform}(\text{unit } d\text{-ball}). \tag{15}$$

For coordinate boxes, integrate the first d barycentric coordinates in their component bounds; the last is one minus their sum. The coordinate-chart Hausdorff factor $\sqrt{K}$ divided by simplex volume leaves factor $d!$. For TV, $\frac{1}{2}\|y - m\|_1 \leq r$ implies $|y_j - m_j| \leq r$ because positive and negative differences each sum in magnitude to TV. This supplies a bounding box in the first d coordinates; retain exact simplex/TV membership inside it. In $K = 3$, an untruncated interior TV ball has normalized area $6r^2$, providing an analytic implementation check.

ILR and smoothing. For $w = \mathrm{softmax}(Bz)$, the tangent-space Jacobian is $K\prod_j w_j$. To see this, the derivative in the tangent space is represented by $B^T(\mathrm{diag}\, w - ww^T)B$; its determinant is the product of the nonzero eigenvalues of $\mathrm{diag}\, w - ww^T$. Each principal cofactor of that matrix is $\prod_j w_j$, by the determinant lemma, and the sum of the K cofactors is that eigenvalue product. Hence, for Z uniform on the radius-r ball around $z(m)$,

$$a_{\mathrm{ILR}} = \sqrt{K}\, d! V_d r^d \, \mathbb{E}_Z \left[\prod_j \mathrm{softmax}(BZ)_j \right]. \tag{16}$$

With smoothing, center the ball at $z(T_\epsilon(m))$, write $w = \mathrm{softmax}(BZ)$, multiply the integrand by $(1+K\epsilon)^d$, and retain only $w_j \geq \epsilon/(1+K\epsilon)$ for every j. This follows from $y = (1+K\epsilon)w - \epsilon\mathbf{1}$ and computes the original target region's area, including its clipping at the simplex boundary. Boundary faces themselves have zero full-dimensional area.

Hellinger. Set $s = \sqrt{y}$ and $a = \sqrt{m}$, so both lie on the unit sphere in the positive orthant. The score constraint is $s^T a \geq 1 - r^2$. The map $s \mapsto y = s^2$ has surface-to-simplex Jacobian $2^d\sqrt{K}\prod_j s_j$. Indeed, its squared tangent Gram determinant is $4^d K \prod_j s_j^2$: apply the cofactor identity to the diagonal squared derivative and use $\sum_j s_j^2 = 1$. Division by simplex volume gives normalized Jacobian $2^d d! \prod_j s_j$. Integrate this over the spherical cap, rejecting negative s_j. For $0 < r < 1$, cap

sampling uses $\sin^2\theta \sim \mathrm{Beta}(d/2, 1/2)$ truncated at $1-(1-r^2)^2$. If $r \geq 1$ the entire simplex is covered. Infinite thresholds give area exactly one for every implemented family; finite radii can also cover the whole simplex in some geometries.

Paired refinement and allowance. The audit uses confirmatory repeats $0, 25, 50, 75$ in every $K = 3$ cell and the first four selected locations per group. At a matched location for methods A, B, let D_{main} be the main-resolution area difference and D_s, $s = 1, \ldots, 4$, the higher-resolution differences, pairing scrambles within each method pair. Define

$$e = \max\left\{.001, \left|\overline{D} - D_{\mathrm{main}}\right| + t_{.975,3}\frac{\mathrm{SD}(D_1, \ldots, D_4)}{2}\right\}. \tag{17}$$

Take the maximum over audited locations/repeats for each cell/group/pair, then the maximum over the two groups for its overall comparison. Enlarge each paired outer-trial interval by this amount on both sides. This empirical sensitivity policy can expose unresolved numerical orderings; its finite audit does not certify all unaudited sets or provide a simultaneous confidence bound. The main outer-trial interval remains a separate statistical uncertainty summary. The exact allowance implementation is `analysis/scripts/numerical_allowance.py`.

D Executed pre-study validation

The following are measured implementation and feasibility diagnostics, separate from the completed confirmatory matrix. They establish neither method rankings nor fitted-composition population coverage.

Calibration variability. With $n = 499$, $\alpha = .1$, $k = 450$ and 2,000 test draws, the saved 10,000-repeat continuous-score simulation gives mean population coverage $.9001285$ and variance $.00017523$, compared with the analytic mean $.9$ and variance $.00017964$ in Equation (10). Empirical finite-test coverage variance is $.00022167$ versus $.00022455$ from Equation (11). Population coverage conditional on a realized calibration set falls below $.9$ in $.4835$ of these draws. This illustrates calibration variability rather than a marginal-guarantee violation. Figure 3 reports the actual empirical CDFs with their separate Monte Carlo uncertainty bands. Its inputs are saved scores/count-derived coverage samples, not fitted-composition study trials.

Area integration and pilot computation. Figure 4 covers 66 fixed TV/L2/Hellinger/ILR cases, including smoothed zero-arm scores. This supersedes the earlier 51-case diagnostic while preserving its historical record. At 2^{14} points, the largest discrepancy of the four-scramble mean from a separate 2^{21}-point direct uniform-simplex reference is $.000856$; at 2^{16} it is $.000115$. The reference also has finite numerical error, and these maxima are not universal error bounds. A separate paired high-resolution audit on saved pilot fits computes 304 matched-location contrasts under Equation (17); its largest empirical allowance is $.002515$. The pilot's two-trial intervals are not interpreted as scientific comparisons.

The final-power pilot executes five $K = 3$ mechanisms with two trials each, TV included, 2^{14} points and 32 area locations per group. It takes 14.105111 seconds and peaks at 180,316 KiB RSS in the supplied CPU environment. This is measured pilot cost, not the final study's cost. The source-pinned pilot metadata and saved rows/checkpoints are in `analysis/results/final_candidate_pilot/`; diagnostic and paired-audit records are under `final_diagnostics/` and `paired_gate/`. The analyst's approximately 47-minute main-run forecast, with a two-hour shared-host allowance, was a planning extrapolation; Appendix E reports actual completed-study costs.

Checks, failures and changes. Thirteen implementation checks pass in the integrated checkout, including quantile/infinity branches, ILR isometry, true-zero original-target membership, absent-test-group aggregation, and independent interior/near-boundary volume checks. A missing $\sqrt{K}$ normalization in a working ILR Jacobian was caught and corrected before pilot computation. Early additive integration seeds overlapped between trial/location indices; the later source-pinned pilot uses separated keys. Exploratory pilot outputs are retained without relabeling them as final-code evidence. One audit launcher failed before computation because a sandbox temporary path was unavailable; the corrected job used the canonical shared config. These changes and failure records are supplied in `analysis/`; no confirmatory outcome was produced by the earlier faulty implementations.

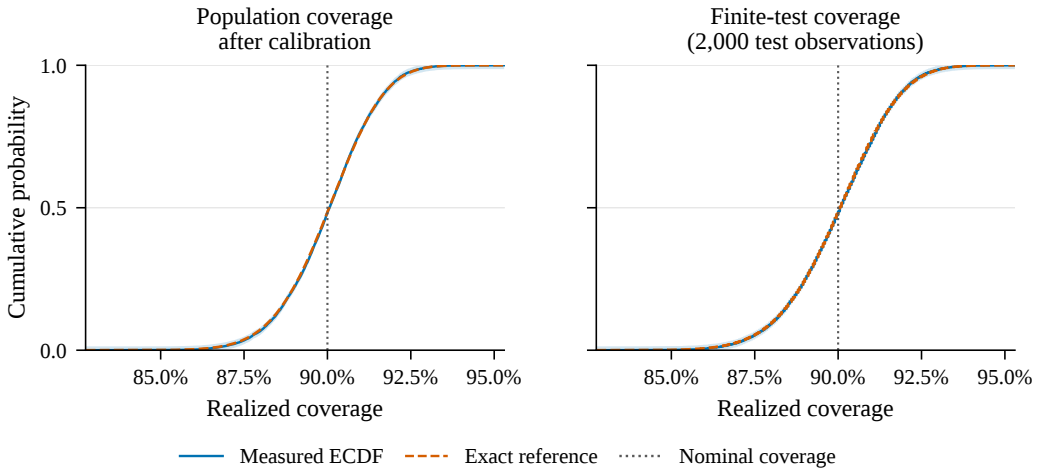


Figure 3: Calibration-conditional coverage varies despite a marginal target, and finite testing adds variation. Each panel uses 10,000 independent calibration realizations of IID Uniform(0,1) scores with $n_{\text{cal}} = 499$, $\alpha = 0.1$ and order-statistic rank 450. Left: actual population coverage at the random threshold versus $\text{Beta}(450, 50)$. Right: covered fraction in 2,000 conditionally IID test draws versus the corresponding beta-binomial CDF. The dotted line is nominal coverage. Shading gives a 95% Dvoretzky–Kiefer–Wolfowitz empirical-CDF band with half-width $\sqrt{\log(2/0.05)/(2R)}$, simultaneous over coverage values separately in each panel. This band describes finite-R Monte Carlo uncertainty of the CDF; the CDF's spread describes calibration/test variability. This fixed continuous-score diagnostic checks the rank reference, not fitted-composition method performance.

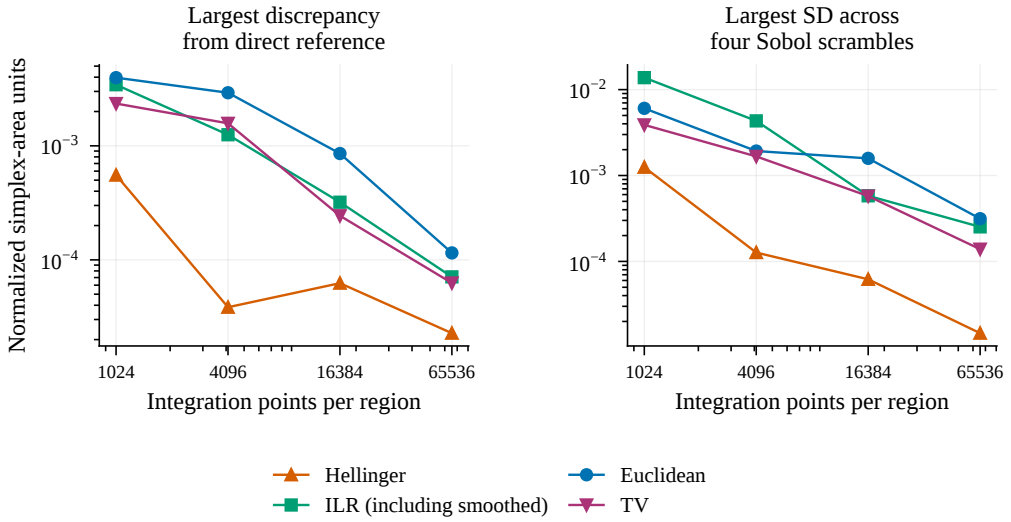


Figure 4: Numerical integration improves with resolution in the checked cases. Curves report maxima within each geometry over 66 fixed center/score/epsilon cases from interior, heterogeneous, boundary, Dirichlet and true-zero mechanisms at $K = 3$. ILR includes its explicitly smoothed zero-arm variants. Left: absolute discrepancy between the mean of four independently scrambled Sobol estimates and an independent 2,097,152-point direct uniform-simplex reference. Right: maximum sample SD across the four scrambles, not a confidence interval or between-trial variability. Both use fraction of original-simplex area. The direct reference also has numerical error; discrepancies are not exact error bounds. These diagnostics do not establish efficiency rankings or precision for geometries absent from the plot.

E Reproduction, resources, source provenance, and impact

Executed evidence and reproduction. The complete frozen configuration and execution record are `analysis/configs/frozen.json` and `analysis/results/confirmatory/execution.json`; exact implementation/configuration hashes are also in the root README. All 19 cells and the scheduled matched-center audit completed. `analysis/results/confirmatory/` retains atomic checkpoints, trial fits/input hashes, raw region-size estimates, summaries, paired intervals, derived diagnostics, numerical allowances, measured cases and the complete inventory audit. Pilots are excluded; retained core displays match the complete final exports.

From the delivery root, `analysis/REPRODUCE.md` and the README give ordinary CPU commands, versions and seeds. Initialize fresh output directories before running both shards, finalization, derived summaries and the `-source-mode files` audit. The original audit also checked its frozen git object. Data are generated locally without downloads or accounts. Renderers read separately hashed original snapshots. The gate in `figures/FRESH_RESULTS.md` checks every consumed scientific input against a specified completed reproduction, including all uncertainty columns and original-order core bootstrap endpoints. Missing or changed inputs refuse rendering; success permits explicitly labeled original-snapshot rendering with unchanged provenance. Pre-study diagnostics remain separate. Script/input maps are in `figures/CORE.md`, `figures/FINAL.md` and their manifests. Building the paper does not run experiments.

Measured resources and historical work. The two scientific jobs, including the full prescribed audit, used 2,401.187 resource CPU seconds and a 1,263.427-second parallel creation-to-end window; main timers were 1,077.780 and 1,154.991 seconds, peak main RSS 181,500 KiB. All 1,900 trials succeeded. Single-thread CPU execution used no GPU or paid API. The scientific snapshot occupies 632,986,948 bytes; packaged size is recorded separately in the root overview. Final aggregation/audit used another 64.342 resource CPU seconds; a subsequent delivery audit/evidence refresh used 7.094. Resource CPU, creation-to-end wall windows and main timers measure different quantities. These measured costs replace the earlier forecast; different hosts vary. Ordinary sequential main runs took 37.2 minutes in total here, plus audits/postprocessing. Pilots, gates and historical failures are additional work recorded in Appendix D. A preconfirmatory launcher failed before any trial; no unfavorable trial was replaced.

Source consultation and independence. The `context/` dossier records actual consultation scope, source revisions, URLs, hashes, failures and exact relay manifests; source PDFs are not redistributed. The positive-domain/isometry claim uses Egozcue et al. [2003]; an unread book and an abstract-only classic support no detailed claims. Bounded consultation establishes neither priority nor an exhaustive review. `analysis/provenance/access-inventory.json` retains all nine positions' dated, attributed access/exposure statements and original retrieval records. Each reports no known prohibited-material encounter; this is an inventory of owner accounts, not independent blanket certification. The integrator's only direct external access was the official generic NeurIPS ethics page. No withheld-paper identity was inferred or searched for.

AI-agent research tools. AI agents contributed source synthesis, design, implementation and manuscript preparation beyond copy editing, affecting scientific choices. Claims rely on executed code, saved measurements and independent review. Responses come from the specified probability distributions.

Impacts, ethics and assets. Separating marginal from group reliability may help detect false reassurance. An adverse use would be treating a nominal guarantee or favorable synthetic result as a deployment safety certificate, especially for poorly covered minorities or changed populations. We delimit target/population claims and distinguish coverage from fairness or decision benefit. These are anticipated implications, not measured downstream effects. There are no human participants, personal datasets or high-risk pretrained models. The NeurIPS Code of Ethics at `https://neurips.cc/public/EthicsGuidelines` was read on 2026-11-03; relevant principles concern representative evaluation, transparency, attribution and reproducibility. Official style assets remain unchanged and attributed in the venue record; no separate archive license grant was supplied, so none is invented. Dependency/font and generated-asset terms are recorded with their locally verified scope in the reproduction package.

F Coordinate and response-slice audits

Figure 5 displays all coordinatewise versus joint core events. The response-defined zero/positive slices differ from the observable covariate groups used for calibration. Paired minority improvements retain their shared-trial uncertainty.

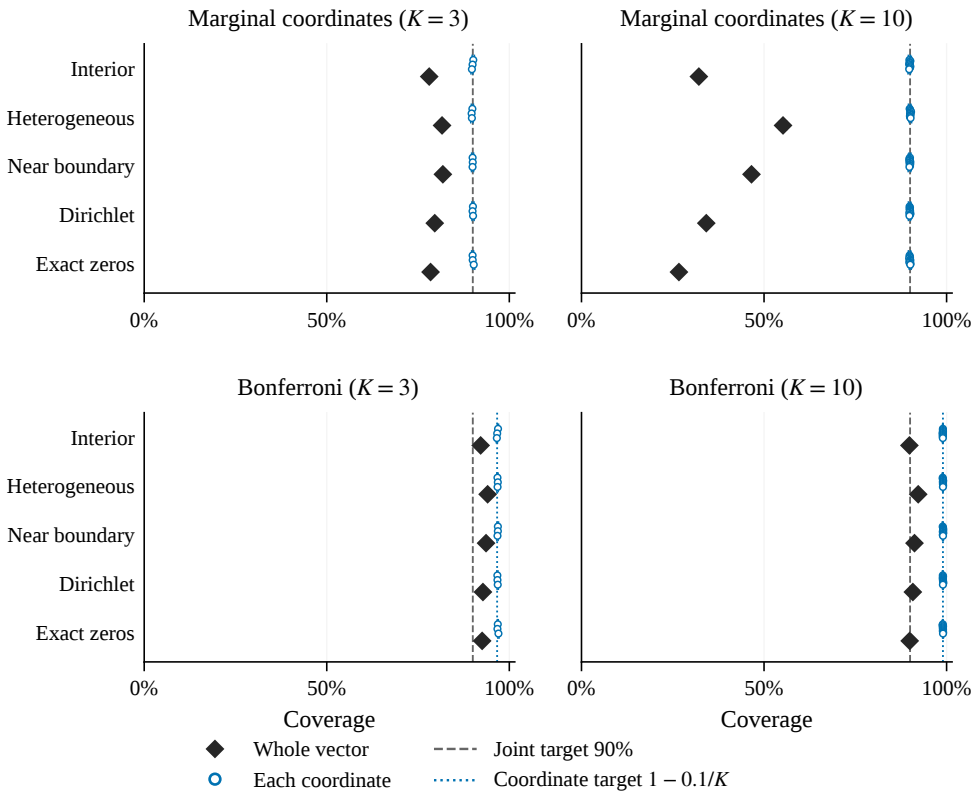


Figure 5: Coordinate coverage and joint coverage are different events. Filled diamonds show measured whole-vector coverage; open circles show each individual coordinate coverage (slight vertical offsets identify components) under marginal-coordinate calibration (top) or Bonferroni correction (bottom). Columns are $K = 3$ and $K = 10$; all five core mechanisms are included. Points show means and bars show pointwise 95% Student-t intervals over 100 independently refitted/regenerated trials per cell, with train/calibration/test counts 2000/499/2000 and shared samples within a trial. Intervals are neither simultaneous nor independent across coordinates. Dashed gray lines mark the 90% joint target; dotted blue lines mark the Bonferroni coordinate target $1 - 0.1/K$. All events concern the original observed composition. The observed joint undercoverage does not contradict its coordinatewise validity.

Table 3: Whole-vector coverage on exact-zero and positive response slices (percent). Entries are mean [pointwise 95% Student-t interval] over the same 100 independent trials per dimension, with train/calibration/test totals 2000/499/2000 and nominal marginal whole-vector coverage 90%. Zero-face means the observed composition has an exact zero; positive means every component is positive. The fitted center and samples are shared across methods. Smoothed ILR uses $\epsilon = 0.001$; unsmoothed ILR is undefined in this arm. These response-defined slices are descriptive audits, not test-time observable groups or conditional guarantees. Their differing coverage can be hidden by the overall near-90% means in the complete table. Full-dimensional area gives faces zero measure and cannot quantify informativeness for the face-supported component.

K	Method	Zero-face coverage	Positive coverage
3	Euclidean	69.26 [68.54, 69.98]	97.15 [96.97, 97.33]
3	TV	69.38 [68.66, 70.10]	97.12 [96.95, 97.30]
3	Hellinger	61.15 [60.14, 62.17]	99.73 [99.69, 99.77]
3	Smoothed ILR	59.85 [58.78, 60.92]	100.00 [100.00, 100.00]
10	Euclidean	85.29 [84.83, 85.76]	91.38 [91.13, 91.63]
10	TV	83.67 [83.19, 84.15]	91.94 [91.69, 92.18]
10	Hellinger	72.25 [71.51, 72.99]	95.89 [95.72, 96.05]
10	Smoothed ILR	59.82 [58.66, 60.99]	100.00 [100.00, 100.00]

Table 4: Paired minority-coverage improvements over global ILR (percentage points). Means [pointwise 95% Student-t intervals] of within-trial method differences over 100 independent trials in the heterogeneous mechanism at each dimension; train/calibration/test counts are 2000/499/2000. Positive values favor the listed method. The observable minority has population prevalence 0.1. Intervals preserve pairing and are not obtained by subtracting marginal interval bounds. Only fixed-group calibration supplies the stated fixed-group guarantee; fitted scaling alone does not.

K	Method	Improvement (pp)
3	Scaled ILR	73.11 [72.09, 74.13]
3	Group ILR	74.35 [73.17, 75.53]
10	Scaled ILR	83.73 [82.31, 85.15]
10	Group ILR	85.84 [84.29, 87.40]

G Measured fitted-region cases

These are all 32 method/case records selected before outcomes: core K 3 trial 0, test observation 0, shared across methods. Contours are approximate displays of the analyst-owned membership rule, not area integration or representative population estimates. The original recorded target's membership and center inclusion were checked for every case. Near-corner marks can overlap; exact coordinates remain in the delivered case file.

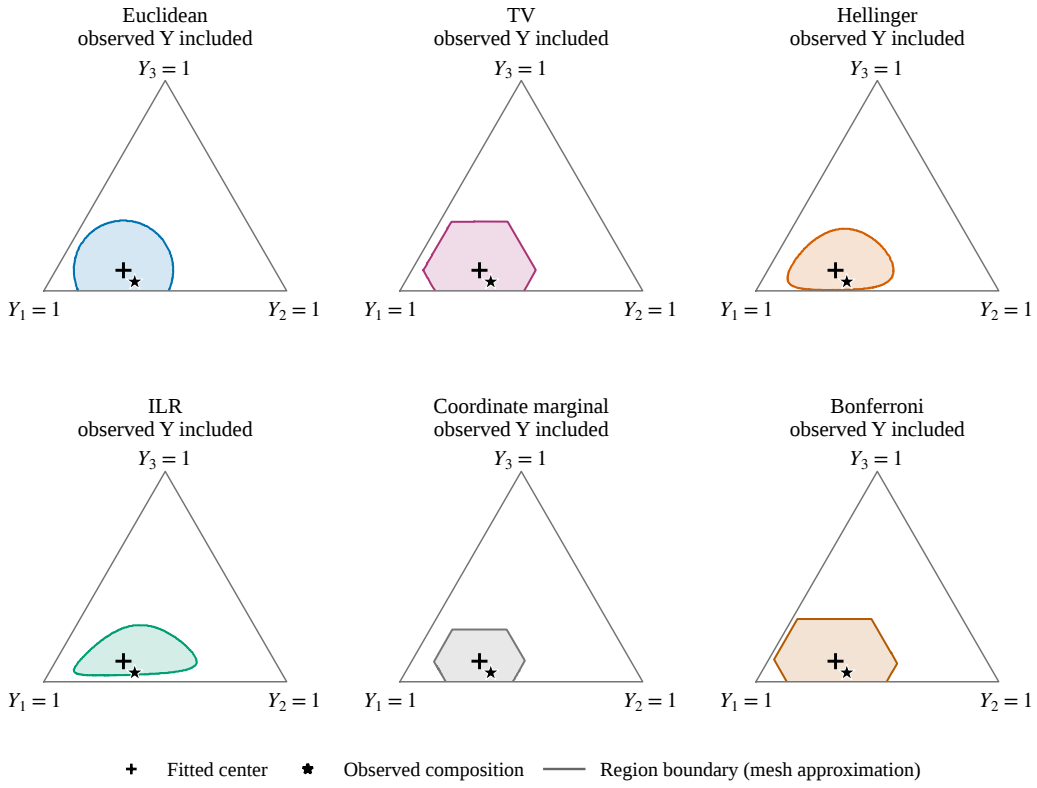


Figure 6: Interior measured fitted regions on the original $K = 3$ simplex. The case is test observation 0 from trial 0, selected before outcomes in every core mechanism and shared across methods, with 2000 training, 499 calibration and 2000 test observations and nominal whole-vector coverage 90%. Plus and star mark the common fitted inverse-ILR center and the actual observed composition; marks can overlap near a vertex. Vertex $Y_j = 1$ has the other two coordinates zero. Shading/lines approximate recorded set membership using regular and center-weighted barycentric meshes (300 and 150 subdivisions), with binary-membership interpolation at 0.5. Titles report exact membership of the original observation, checked with the saved-score evaluator. These contours have display resolution, not inferential error bars or an area interpretation; this single fitted example is not population coverage evidence.

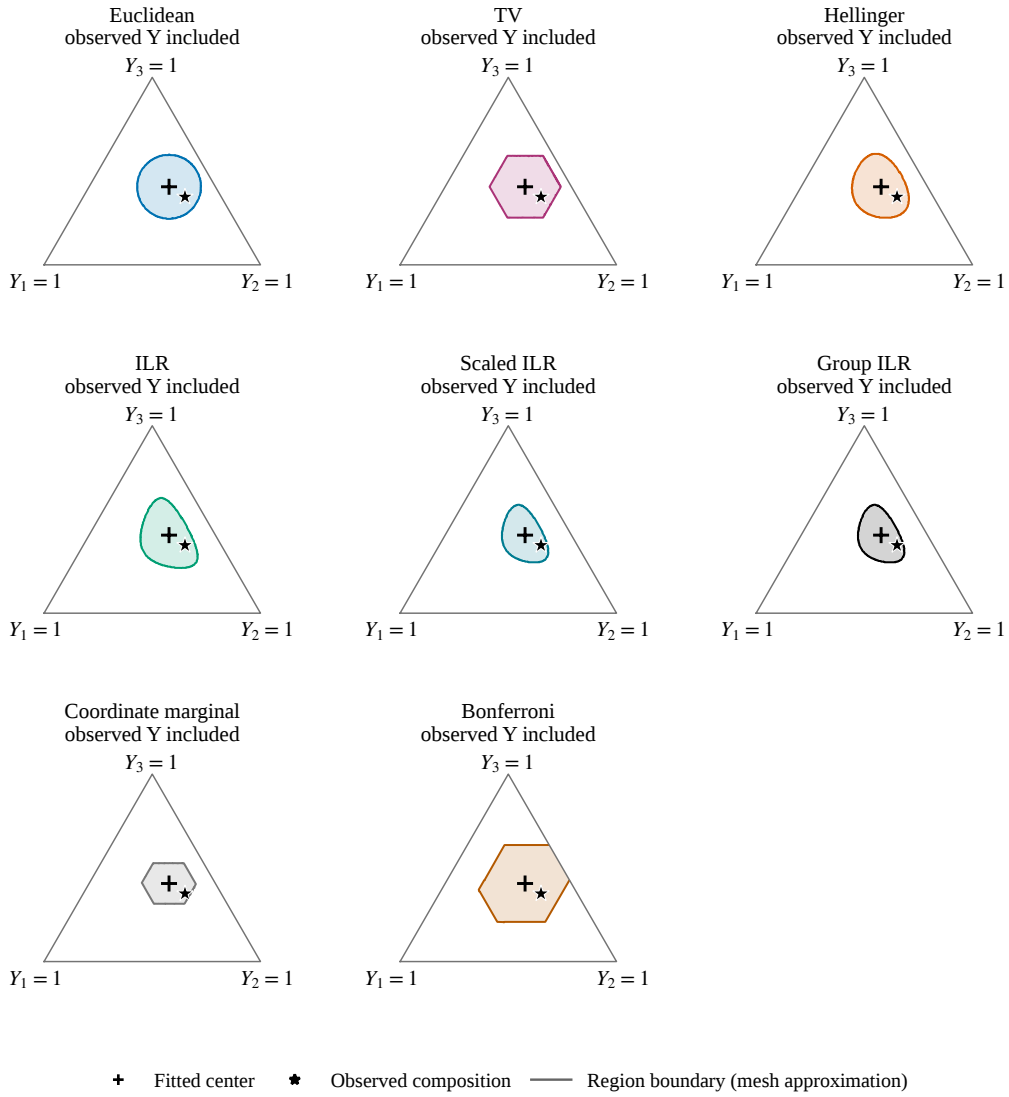


Figure 7: Heterogeneous measured fitted regions on the original $K = 3$ simplex. The case is test observation 0 from trial 0, selected before outcomes in every core mechanism and shared across methods, with 2000 training, 499 calibration and 2000 test observations and nominal whole-vector coverage 90%. Plus and star mark the common fitted inverse-ILR center and the actual observed composition; marks can overlap near a vertex. Vertex $Y_j = 1$ has the other two coordinates zero. Shading/lines approximate recorded set membership using regular and center-weighted barycentric meshes (300 and 150 subdivisions), with binary-membership interpolation at 0.5. Titles report exact membership of the original observation, checked with the saved-score evaluator. These contours have display resolution, not inferential error bars or an area interpretation; this single fitted example is not population coverage evidence.

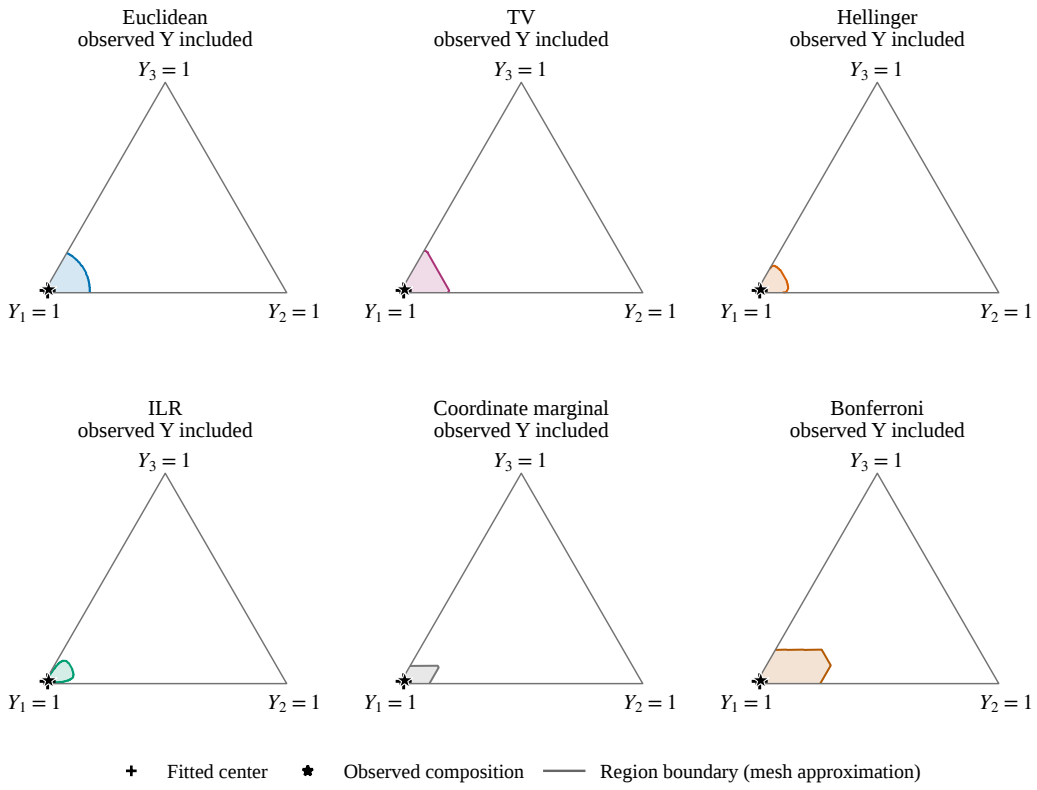


Figure 8: Near-boundary measured fitted regions on the original $K = 3$ simplex. The case is test observation 0 from trial 0, selected before outcomes in every core mechanism and shared across methods, with 2000 training, 499 calibration and 2000 test observations and nominal whole-vector coverage 90%. Plus and star mark the common fitted inverse-ILR center and the actual observed composition; marks can overlap near a vertex. Vertex $Y_j = 1$ has the other two coordinates zero. Shading/lines approximate recorded set membership using regular and center-weighted barycentric meshes (300 and 150 subdivisions), with binary-membership interpolation at 0.5. Titles report exact membership of the original observation, checked with the saved-score evaluator. These contours have display resolution, not inferential error bars or an area interpretation; this single fitted example is not population coverage evidence.

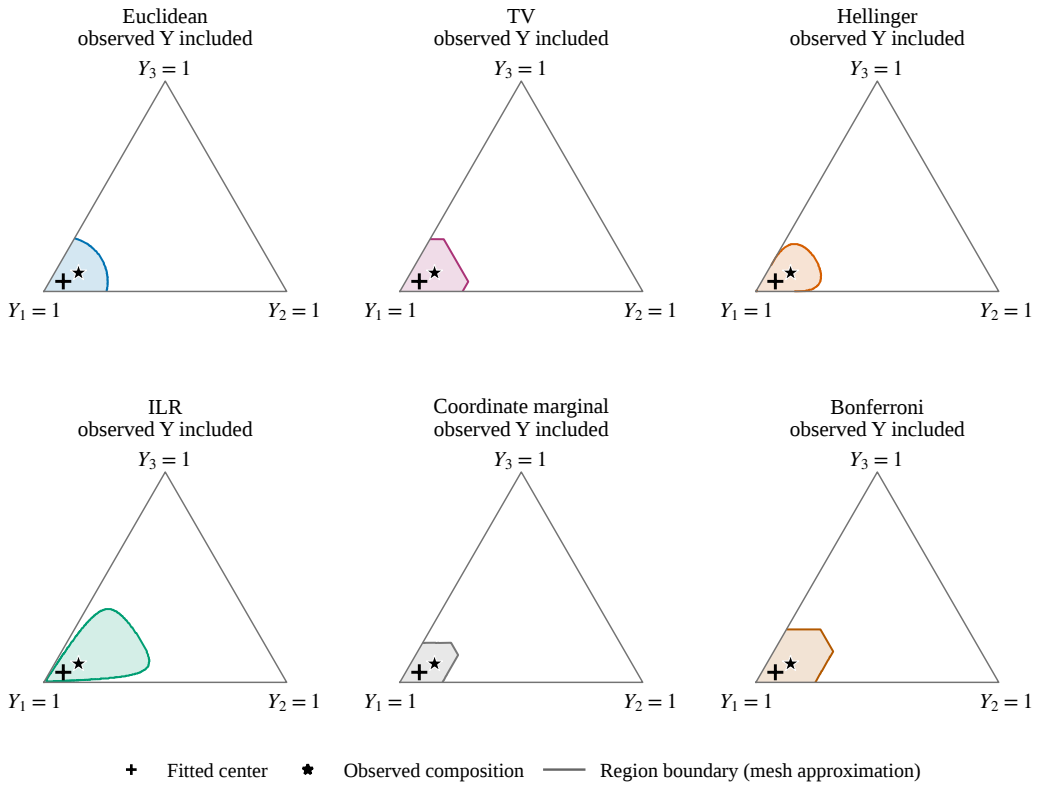


Figure 9: Dirichlet measured fitted regions on the original $K = 3$ simplex. The case is test observation 0 from trial 0, selected before outcomes in every core mechanism and shared across methods, with 2000 training, 499 calibration and 2000 test observations and nominal whole-vector coverage 90%. Plus and star mark the common fitted inverse-ILR center and the actual observed composition; marks can overlap near a vertex. Vertex $Y_j = 1$ has the other two coordinates zero. Shading/lines approximate recorded set membership using regular and center-weighted barycentric meshes (300 and 150 subdivisions), with binary-membership interpolation at 0.5. Titles report exact membership of the original observation, checked with the saved-score evaluator. These contours have display resolution, not inferential error bars or an area interpretation; this single fitted example is not population coverage evidence.

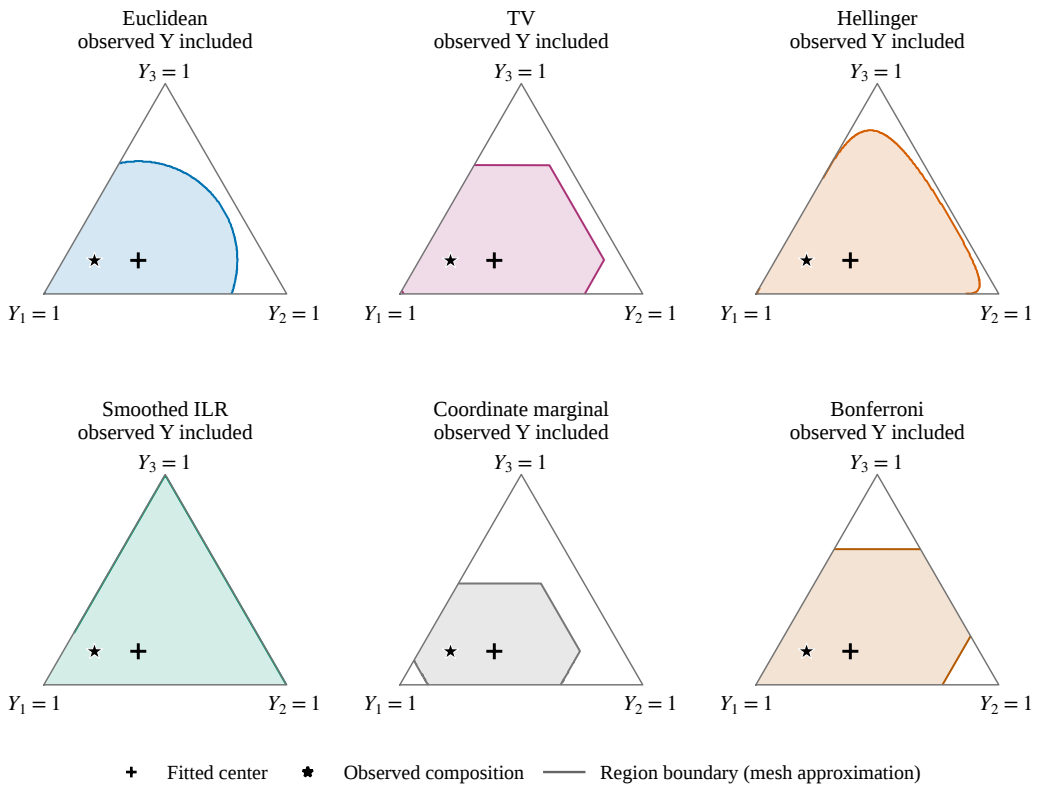


Figure 10: Exact-zero measured fitted regions on the original $K = 3$ simplex. The case is test observation 0 from trial 0, selected before outcomes in every core mechanism and shared across methods, with 2000 training, 499 calibration and 2000 test observations and nominal whole-vector coverage 90%. Plus and star mark the common fitted inverse-ILR center and the actual observed composition; marks can overlap near a vertex. Vertex $Y_j = 1$ has the other two coordinates zero. Shading/lines approximate recorded set membership using regular and center-weighted barycentric meshes (300 and 150 subdivisions), with binary-membership interpolation at 0.5. Titles report exact membership of the original observation, checked with the saved-score evaluator. These contours have display resolution, not inferential error bars or an area interpretation; this single fitted example is not population coverage evidence. Smoothed ILR uses $\epsilon = 0.001$; unsmoothed ILR is undefined at exact zeros and is omitted here.

H Numerical contrasts and sensitivity results

The plots and tables below retain enlarged paired area uncertainty, the specific prevalence-shift comparison and the three declared sensitivity families. Cross-cell area contrasts have no within-cell numerical ranking certification. The complete machine-readable paired-contrast and allowance files preserve all comparisons, including unresolved outcomes.

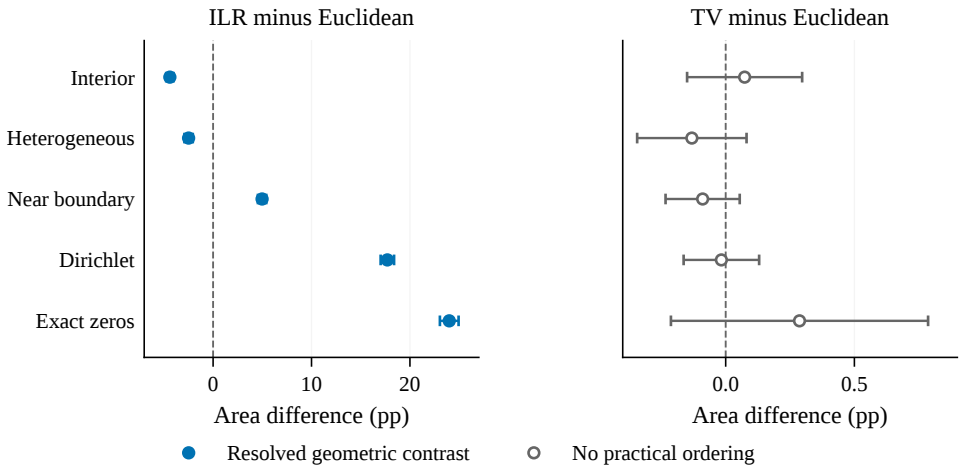


Figure 11: Paired numerical resolution of principal $K = 3$ area contrasts. Points show within-trial mean differences (percentage points of full original-simplex area); bars are pointwise 95% paired Student-t intervals enlarged on each side by the supplied empirical numerical allowance. The audit uses all 14 $K = 3$ cells, repeats 0/25/50/75, first four test centers per observed group, power16 and four independent scrambles, matched to the power14 main estimates. The allowance is the maximum prescribed matched-center refinement discrepancy plus paired-scramble term, with floor .001; it is finite sensitivity evidence, not a uniform bound. Filled marks pass the frozen sign and .005 area practical margin; open marks do not. Panels use different labeled scales. ILR is smoothed with $\epsilon = .001$ in the exact-zero arm; faces have zero full-dimensional area. Interpret geometry at achieved relevant coverage, not as universal superiority.

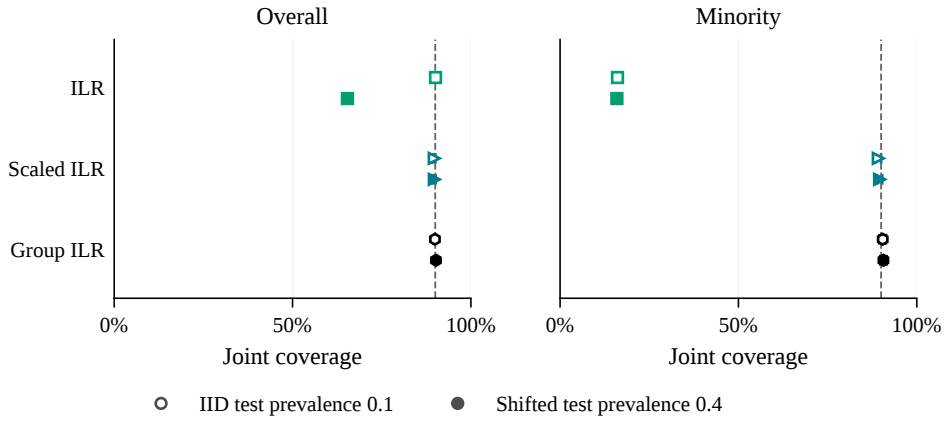


Figure 12: Observable-group prevalence shift at $K = 3$. Open marks are IID test prevalence .1; filled marks change only minority prevalence to .4 with invariant within-group laws. Procedures share training/calibration data within each repeat. Means and pointwise 95% Student-t intervals use 100 independent regenerated/refitted trials per arm, train/calibration/test 2000/499/2000; the shifted test arrays differ despite matched random-number seeds. The dashed line marks 90% joint observed-composition coverage. Global calibration loses marginal protection in this arm; fixed-group validity applies to this particular mixture shift. Scaling is not a general shift remedy. Paired population effects and full denominators are retained in the companion tables/data.

Table 5: Coverage under the specified observable-group prevalence shift (percent). Means [pointwise 95% Student-t intervals] over 100 independent $K = 3$ trials, each with train/calibration/test counts 2000/499/2000. Test minority prevalence changes from .1 to .4 while within-group laws remain fixed. This is the particular shift for which fixed-group validity survives; neither fitted scaling nor this test establishes general shift robustness.

Method	Overall	Minority
ILR	65.39 [64.99, 65.80]	15.92 [15.25, 16.60]
Scaled ILR	89.74 [89.38, 90.10]	89.57 [88.82, 90.31]
Group ILR	90.20 [89.82, 90.59]	90.65 [89.78, 91.52]

Table 6: Paired shift-minus-IID changes in population coverage (percentage points). Means [pointwise 95% Student-t intervals] of within-repeat effects over 100 trials, preserving common random-number pairing. Training/calibration samples are shared; shifted test arrays differ. This is a comparison of the declared prevalence arm, not independent datasets or arbitrary shifts.

Method	Shift minus IID (pp)
ILR	-24.69 [-24.93, -24.45]
Scaled ILR	0.01 [-0.21, 0.23]
Group ILR	0.29 [0.02, 0.56]

Table 7: Observed calibration-size/nominal sensitivity for interior $K = 3$ ILR (percent and percentage points). Coverage entries are mean [pointwise 95% Student-t interval] over 100 independent regenerated/refitted trials per cell; the last column is the across-trial standard deviation, not uncertainty of the mean. Train/test counts are 2000/2000. Sensitivity arms share repeat seeds; no cross-cell independence or universal monotonic relation is asserted.

n_{cal}	Target (%)	Coverage (%)	Trial SD (pp)
99	90	89.39 [88.77, 90.00]	3.09
499	90	89.87 [89.56, 90.18]	1.56
1999	90	89.90 [89.69, 90.11]	1.06
499	95	94.97 [94.75, 95.19]	1.12

Table 8: Minority calibration cost for fixed-group ILR at $K = 3$ (percent). Means [pointwise 95% Student-t intervals] over 100 independent trials, with 2000 training/test observations and the listed calibration total. The observable minority has prevalence .1. At calibration total 99, 31 of 100 minority calibrations produce an infinite threshold, retaining the full valid domain. In these selected cells the separately supplied exact full-domain rate equals the infinite-threshold rate; the two metrics are generally distinct. A full-domain case is retained conservatism, not a failed or dropped trial.

n_{cal}	Target (%)	Minority coverage	Infinite threshold
99	90	94.93 [93.64, 96.22]	31.00 [21.78, 40.22]
499	90	90.44 [89.55, 91.34]	0.00 [0.00, 0.00]
1999	90	89.43 [88.82, 90.03]	0.00 [0.00, 0.00]
499	95	95.53 [94.90, 96.16]	0.00 [0.00, 0.00]

Table 9: Explicitly smoothed ILR on the actual-zero $K = 3$ mechanism. Coverage is percent; area is a normalized original-simplex fraction. Means [pointwise 95% Student-t intervals] use 100 independent trials per cell (train/calibration/test 2000/499/2000), with the common predictor held fixed across the paired epsilon arms. The tiny-epsilon estimate above one is retained numerical error, not true area above one or a full-domain certificate; the analysis retains 159 trial/group range warnings. Near-full area cannot describe coverage on zero faces, which have zero full-dimensional measure. Area intervals here are outer-trial only; cross-cell epsilon differences have no cross-cell numerical ranking certification. Zero-face coverage is a descriptive response slice with no test-time conditional guarantee.

ϵ	Overall coverage	Zero-face coverage	Area fraction
10^{-6}	89.95 [89.67, 90.22]	59.84 [58.75, 60.93]	1.000018 [0.999994, 1.000041]
10^{-3}	89.95 [89.68, 90.22]	59.85 [58.78, 60.92]	0.997125 [0.996953, 0.997297]
10^{-2}	89.99 [89.72, 90.27]	60.03 [58.94, 61.12]	0.968920 [0.967228, 0.970613]

I Complete study tables and domain accounting

All 446 declared method/group entries are retained: 426 defined and 20 explicit unsmoothed-ILR inapplicability entries, each with 100 recorded trials. Counts, actual nonempty denominators, fit/calibration counts and numerical status remain in the exact companion CSV. Zero-count groups are not assigned zero coverage; undefined positive-weight area propagates undefined. Infinite thresholds and full-domain identification bounds/unknown status are distinct diagnostics. These are descriptive pointwise intervals, not simultaneous acceptance tests.

Table 10: Complete whole-vector coverage (percent) and original-simplex area (normalized fraction). Entries are means [95% Student-t confidence intervals] over independent refitted trials; R_c/R gives nonempty coverage trials / recorded trials (100 per cell). Training/test totals are 2000/2000; calibration totals and nominal levels are in each heading. Area is unavailable at $K = 10$ and for response-defined zero/positive slices. At $K = 3$, area averages stratified test-X locations using observed test-group prevalence. These area intervals describe outer-trial uncertainty; paired rankings additionally require the empirical numerical allowance in the contrast file. † marks minority mean-coverage interval halfwidth > 2 percentage points. Zero-face/positive groups are descriptive response slices, not test-time guarantees. No high-dimensional area ordering is reported. Numerical estimates near one are retained untruncated; the tiny-epsilon arm includes an estimate above one, which is numerical error rather than true area above one or a full-domain certificate.

Method	Joint coverage (%)	Area fraction	R_c/R
interior_k3; all; K=3, n_cal=499, 1-alpha=0.9			
Euclidean	89.91 [89.61, 90.20]	0.271686 [0.268627, 0.274745]	100/100
Total variation	89.86 [89.59, 90.14]	0.272422 [0.269430, 0.275414]	100/100
Hellinger	89.93 [89.63, 90.23]	0.229529 [0.226781, 0.232278]	100/100
ILR	89.87 [89.56, 90.18]	0.227746 [0.224784, 0.230708]	100/100
Coordinate marginal	78.06 [77.66, 78.47]	0.177814 [0.175832, 0.179796]	100/100
Bonferroni	92.15 [91.88, 92.42]	0.299971 [0.296306, 0.303636]	100/100
interior_k10; all; K=10, n_cal=499, 1-alpha=0.9			
Euclidean	89.67 [89.36, 89.99]	—	100/100
Total variation	90.05 [89.76, 90.35]	—	100/100
Hellinger	89.86 [89.56, 90.16]	—	100/100
ILR	89.91 [89.59, 90.23]	—	100/100
Coordinate marginal	32.15 [31.76, 32.54]	—	100/100
Bonferroni	89.81 [89.50, 90.13]	—	100/100
heterogeneous_k3; all; K=3, n_cal=499, 1-alpha=0.9			
Euclidean	89.93 [89.60, 90.25]	0.164309 [0.159679, 0.168938]	100/100
Total variation	89.81 [89.49, 90.14]	0.162994 [0.158520, 0.167469]	100/100
Hellinger	89.89 [89.60, 90.19]	0.139732 [0.135367, 0.144097]	100/100
ILR	90.08 [89.80, 90.36]	0.139416 [0.134011, 0.144821]	100/100
Scaled ILR	89.73 [89.44, 90.02]	0.151633 [0.150037, 0.153228]	100/100
Group ILR	89.91 [89.61, 90.21]	0.153259 [0.151560, 0.154959]	100/100
Coordinate marginal	81.58 [81.11, 82.05]	0.093349 [0.091324, 0.095374]	100/100
Bonferroni	94.06 [93.84, 94.28]	0.296867 [0.285784, 0.307950]	100/100

Table 10 (continued)

Method	Joint coverage (%)	Area fraction	R_c/R
heterogeneous_k10; all; K=10, n_cal=499, 1-alpha=0.9			
Euclidean	90.08 [89.80, 90.35]	—	100/100
Total variation	90.01 [89.73, 90.29]	—	100/100
Hellinger	90.05 [89.78, 90.32]	—	100/100
ILR	90.03 [89.75, 90.30]	—	100/100
Scaled ILR	89.82 [89.54, 90.10]	—	100/100
Group ILR	90.03 [89.76, 90.30]	—	100/100
Coordinate marginal	55.20 [54.43, 55.97]	—	100/100
Bonferroni	92.23 [92.07, 92.38]	—	100/100
boundary_k3; all; K=3, n_cal=499, 1-alpha=0.9			
Euclidean	89.94 [89.66, 90.21]	0.073809 [0.071667, 0.075951]	100/100
Total variation	89.95 [89.67, 90.23]	0.072914 [0.070867, 0.074960]	100/100
Hellinger	89.94 [89.64, 90.23]	0.067061 [0.065101, 0.069021]	100/100
ILR	89.90 [89.60, 90.20]	0.123582 [0.119554, 0.127611]	100/100
Coordinate marginal	81.80 [81.46, 82.15]	0.038991 [0.038031, 0.039951]	100/100
Bonferroni	93.63 [93.40, 93.85]	0.117775 [0.114311, 0.121238]	100/100
boundary_k10; all; K=10, n_cal=499, 1-alpha=0.9			
Euclidean	90.02 [89.71, 90.32]	—	100/100
Total variation	90.00 [89.69, 90.31]	—	100/100
Hellinger	90.04 [89.74, 90.34]	—	100/100
ILR	89.95 [89.71, 90.19]	—	100/100
Coordinate marginal	46.56 [46.20, 46.93]	—	100/100
Bonferroni	91.21 [90.93, 91.49]	—	100/100
dirichlet_k3; all; K=3, n_cal=499, 1-alpha=0.9			
Euclidean	89.86 [89.56, 90.16]	0.113017 [0.111358, 0.114675]	100/100
Total variation	89.96 [89.67, 90.25]	0.112850 [0.111258, 0.114442]	100/100
Hellinger	89.84 [89.55, 90.13]	0.114384 [0.112497, 0.116272]	100/100
ILR	90.05 [89.78, 90.31]	0.290070 [0.283622, 0.296519]	100/100
Coordinate marginal	79.58 [79.20, 79.97]	0.078056 [0.076874, 0.079237]	100/100
Bonferroni	92.78 [92.53, 93.02]	0.134295 [0.132301, 0.136288]	100/100
dirichlet_k10; all; K=10, n_cal=499, 1-alpha=0.9			
Euclidean	89.92 [89.59, 90.25]	—	100/100
Total variation	89.81 [89.47, 90.14]	—	100/100
Hellinger	89.97 [89.65, 90.28]	—	100/100
ILR	90.04 [89.76, 90.31]	—	100/100
Coordinate marginal	34.18 [33.76, 34.60]	—	100/100
Bonferroni	90.76 [90.45, 91.07]	—	100/100

Table 10 (continued)

Method	Joint coverage (%)	Area fraction	R_c/R
zeros_k3; all; K=3, n_cal=499, 1-alpha=0.9, epsilon=0.001			
Euclidean	90.17 [89.88, 90.46]	0.757191 [0.750496, 0.763887]	100/100
Total variation	90.18 [89.89, 90.47]	0.760059 [0.753674, 0.766444]	100/100
Hellinger	90.07 [89.80, 90.35]	0.890406 [0.885346, 0.895466]	100/100
Smoothed ILR	89.95 [89.68, 90.22]	0.997125 [0.996953, 0.997297]	100/100
Coordinate marginal	78.43 [78.05, 78.82]	0.567251 [0.561118, 0.573383]	100/100
Bonferroni	92.56 [92.33, 92.79]	0.820360 [0.814554, 0.826167]	100/100
ILR (undefined)	Undefined at exact zeros		
zeros_k10; all; K=10, n_cal=499, 1-alpha=0.9, epsilon=0.001			
Euclidean	89.86 [89.58, 90.13]	—	100/100
Total variation	89.87 [89.59, 90.14]	—	100/100
Hellinger	89.97 [89.68, 90.26]	—	100/100
Smoothed ILR	89.95 [89.64, 90.25]	—	100/100
Coordinate marginal	26.69 [26.34, 27.04]	—	100/100
Bonferroni	89.90 [89.60, 90.20]	—	100/100
ILR (undefined)	Undefined at exact zeros		
interior_ncal99; all; K=3, n_cal=99, 1-alpha=0.9			
Euclidean	89.69 [89.05, 90.33]	0.273781 [0.266353, 0.281209]	100/100
Total variation	89.45 [88.83, 90.07]	0.272054 [0.264908, 0.279199]	100/100
Hellinger	89.80 [89.20, 90.40]	0.232230 [0.226205, 0.238254]	100/100
ILR	89.39 [88.77, 90.00]	0.225966 [0.220367, 0.231564]	100/100
Coordinate marginal	77.63 [76.90, 78.36]	0.177947 [0.173884, 0.182009]	100/100
Bonferroni	92.12 [91.60, 92.65]	0.308341 [0.300625, 0.316057]	100/100
interior_ncal1999; all; K=3, n_cal=1999, 1-alpha=0.9			
Euclidean	90.01 [89.80, 90.22]	0.271903 [0.269903, 0.273902]	100/100
Total variation	89.90 [89.69, 90.11]	0.272454 [0.270606, 0.274303]	100/100
Hellinger	89.99 [89.78, 90.20]	0.229849 [0.228001, 0.231697]	100/100
ILR	89.90 [89.69, 90.11]	0.227648 [0.225494, 0.229801]	100/100
Coordinate marginal	78.25 [77.97, 78.53]	0.178592 [0.177440, 0.179745]	100/100
Bonferroni	91.98 [91.77, 92.19]	0.296582 [0.294393, 0.298771]	100/100
interior_alpha05; all; K=3, n_cal=499, 1-alpha=0.95			
Euclidean	94.88 [94.64, 95.12]	0.347920 [0.343100, 0.352740]	100/100
Total variation	94.87 [94.63, 95.12]	0.347546 [0.342900, 0.352192]	100/100
Hellinger	94.89 [94.64, 95.13]	0.294643 [0.290506, 0.298780]	100/100
ILR	94.97 [94.75, 95.19]	0.285704 [0.281930, 0.289478]	100/100
Coordinate marginal	88.17 [87.87, 88.48]	0.252949 [0.250120, 0.255779]	100/100
Bonferroni	95.88 [95.69, 96.07]	0.369496 [0.364641, 0.374351]	100/100

Table 10 (continued)

Method	Joint coverage (%)	Area fraction	R_c/R
heterogeneous_ncal99; all; K=3, n_cal=99, 1-alpha=0.9			
Euclidean	90.37 [89.80, 90.95]	0.187456 [0.175235, 0.199678]	100/100
Total variation	90.47 [89.92, 91.01]	0.190132 [0.177699, 0.202566]	100/100
Hellinger	90.43 [89.85, 91.01]	0.178315 [0.161799, 0.194831]	100/100
ILR	90.46 [89.91, 91.02]	0.189729 [0.168092, 0.211366]	100/100
Scaled ILR	89.65 [89.04, 90.27]	0.152768 [0.150147, 0.155390]	100/100
Group ILR	90.55 [89.99, 91.12]	0.162084 [0.159171, 0.164998]	100/100
Coordinate marginal	82.19 [81.27, 83.11]	0.102987 [0.097501, 0.108472]	100/100
Bonferroni	94.42 [93.98, 94.86]	0.358066 [0.333708, 0.382424]	100/100
heterogeneous_ncal1999; all; K=3, n_cal=1999, 1-alpha=0.9			
Euclidean	89.94 [89.78, 90.10]	0.161159 [0.159105, 0.163212]	100/100
Total variation	89.98 [89.82, 90.13]	0.162343 [0.160316, 0.164370]	100/100
Hellinger	90.01 [89.86, 90.16]	0.138045 [0.135906, 0.140183]	100/100
ILR	90.03 [89.89, 90.18]	0.133698 [0.131312, 0.136083]	100/100
Scaled ILR	89.76 [89.57, 89.96]	0.151493 [0.150049, 0.152937]	100/100
Group ILR	89.78 [89.59, 89.98]	0.151660 [0.150328, 0.152992]	100/100
Coordinate marginal	82.03 [81.80, 82.27]	0.094087 [0.093110, 0.095064]	100/100
Bonferroni	93.99 [93.87, 94.10]	0.280525 [0.275425, 0.285625]	100/100
heterogeneous_alpha05; all; K=3, n_cal=499, 1-alpha=0.95			
Euclidean	95.06 [94.84, 95.28]	0.362591 [0.347105, 0.378077]	100/100
Total variation	95.03 [94.81, 95.24]	0.363209 [0.347920, 0.378498]	100/100
Hellinger	95.02 [94.81, 95.24]	0.384012 [0.368148, 0.399875]	100/100
ILR	94.96 [94.76, 95.16]	0.446448 [0.429566, 0.463330]	100/100
Scaled ILR	94.76 [94.54, 94.97]	0.177479 [0.175521, 0.179437]	100/100
Group ILR	94.97 [94.76, 95.17]	0.179777 [0.177838, 0.181716]	100/100
Coordinate marginal	91.24 [90.93, 91.56]	0.191591 [0.185249, 0.197932]	100/100
Bonferroni	96.63 [96.46, 96.80]	0.512207 [0.496126, 0.528289]	100/100

Table 10 (continued)

Method	Joint coverage (%)	Area fraction	R_c/R
heterogeneous_shift; all; K=3, n_cal=499, 1-alpha=0.9, group-prevalence shift 0.10 to 0.40			
		0.162405	
Euclidean	68.39 [67.94, 68.83]	[0.157892, 0.166917]	100/100
		0.161083	
Total variation	68.20 [67.76, 68.64]	[0.156725, 0.165440]	100/100
		0.137602	
Hellinger	66.14 [65.74, 66.54]	[0.133330, 0.141874]	100/100
		0.136383	
ILR	65.39 [64.99, 65.80]	[0.131124, 0.141642]	100/100
		0.375446	
Scaled ILR	89.74 [89.38, 90.10]	[0.371285, 0.379607]	100/100
		0.381066	
Group ILR	90.20 [89.82, 90.59]	[0.376475, 0.385657]	100/100
		0.092978	
Coordinate marginal	59.54 [59.11, 59.96]	[0.090980, 0.094976]	100/100
		0.293238	
Bonferroni	77.58 [76.98, 78.18]	[0.282271, 0.304206]	100/100
zeros_eps1e-06; all; K=3, n_cal=499, 1-alpha=0.9, epsilon=1e-06			
		0.757191	
Euclidean	90.17 [89.88, 90.46]	[0.750496, 0.763887]	100/100
		0.760059	
Total variation	90.18 [89.89, 90.47]	[0.753674, 0.766444]	100/100
		0.890406	
Hellinger	90.07 [89.80, 90.35]	[0.885346, 0.895466]	100/100
		1.000018	
Smoothed ILR	89.95 [89.67, 90.22]	[0.999994, 1.000041]	100/100
		0.567251	
Coordinate marginal	78.43 [78.05, 78.82]	[0.561118, 0.573383]	100/100
		0.820360	
Bonferroni	92.56 [92.33, 92.79]	[0.814554, 0.826167]	100/100
ILR (undefined)	Undefined at exact zeros		
zeros_eps0.01; all; K=3, n_cal=499, 1-alpha=0.9, epsilon=0.01			
		0.757191	
Euclidean	90.17 [89.88, 90.46]	[0.750496, 0.763887]	100/100
		0.760059	
Total variation	90.18 [89.89, 90.47]	[0.753674, 0.766444]	100/100
		0.890406	
Hellinger	90.07 [89.80, 90.35]	[0.885346, 0.895466]	100/100
		0.968920	
Smoothed ILR	89.99 [89.72, 90.27]	[0.967228, 0.970613]	100/100
		0.567251	
Coordinate marginal	78.43 [78.05, 78.82]	[0.561118, 0.573383]	100/100
		0.820360	
Bonferroni	92.56 [92.33, 92.79]	[0.814554, 0.826167]	100/100
ILR (undefined)	Undefined at exact zeros		

Table 11: Complete whole-vector coverage (percent) and original-simplex area (normalized fraction). Entries are means [95% Student-t confidence intervals] over independent refitted trials; R_c/R gives nonempty coverage trials / recorded trials (100 per cell). Training/test totals are 2000/2000; calibration totals and nominal levels are in each heading. Area is unavailable at $K = 10$ and for response-defined zero/positive slices. At $K = 3$, area averages stratified test-X locations using observed test-group prevalence. These area intervals describe outer-trial uncertainty; paired rankings additionally require the empirical numerical allowance in the contrast file. † marks minority mean-coverage interval halfwidth > 2 percentage points. Zero-face/positive groups are descriptive response slices, not test-time guarantees. No high-dimensional area ordering is reported. Numerical estimates near one are retained untruncated; the tiny-epsilon arm includes an estimate above one, which is numerical error rather than true area above one or a full-domain certificate.

Method	Joint coverage (%)	Area fraction	R_c/R
interior_k3; majority; K=3, n_cal=499, 1-alpha=0.9			
Euclidean	89.85 [89.55, 90.15]	0.272803 [0.269737, 0.275868]	100/100
Total variation	89.80 [89.51, 90.08]	0.273505 [0.270497, 0.276513]	100/100
Hellinger	89.87 [89.57, 90.17]	0.230502 [0.227729, 0.233275]	100/100
ILR	89.85 [89.54, 90.17]	0.229027 [0.225996, 0.232057]	100/100
Coordinate marginal	77.88 [77.47, 78.29]	0.178436 [0.176452, 0.180420]	100/100
Bonferroni	92.09 [91.81, 92.36]	0.301068 [0.297388, 0.304749]	100/100
interior_k3; minority; K=3, n_cal=499, 1-alpha=0.9			
Euclidean	90.44 [89.93, 90.95]	0.261531 [0.258079, 0.264983]	100/100
Total variation	90.46 [89.97, 90.96]	0.262589 [0.259231, 0.265948]	100/100
Hellinger	90.50 [89.99, 91.02]	0.220692 [0.217556, 0.223827]	100/100
ILR	89.98 [89.47, 90.49]	0.216100 [0.212654, 0.219545]	100/100
Coordinate marginal	79.74 [79.01, 80.46]	0.172141 [0.170091, 0.174191]	100/100
Bonferroni	92.71 [92.30, 93.12]	0.290002 [0.285909, 0.294096]	100/100
interior_k10; majority; K=10, n_cal=499, 1-alpha=0.9			
Euclidean	89.79 [89.47, 90.11]	—	100/100
Total variation	90.09 [89.80, 90.39]	—	100/100
Hellinger	89.91 [89.61, 90.21]	—	100/100
ILR	89.94 [89.62, 90.25]	—	100/100
Coordinate marginal	32.18 [31.79, 32.58]	—	100/100
Bonferroni	89.86 [89.54, 90.19]	—	100/100
interior_k10; minority; K=10, n_cal=499, 1-alpha=0.9			
Euclidean	88.61 [88.05, 89.17]	—	100/100
Total variation	89.70 [89.16, 90.24]	—	100/100
Hellinger	89.44 [88.91, 89.97]	—	100/100
ILR	89.68 [89.10, 90.26]	—	100/100
Coordinate marginal	31.85 [31.11, 32.60]	—	100/100
Bonferroni	89.36 [88.92, 89.80]	—	100/100

Table 11 (continued)

Method	Joint coverage (%)	Area fraction	R_c/R
heterogeneous_k3; majority; K=3, n_cal=499, 1-alpha=0.9			
Euclidean	97.09 [96.82, 97.37]	0.164859 [0.160211, 0.169508]	100/100
Total variation	96.99 [96.70, 97.28]	0.163550 [0.159057, 0.168043]	100/100
Hellinger	97.80 [97.54, 98.06]	0.140301 [0.135938, 0.144664]	100/100
ILR	98.25 [98.02, 98.48]	0.140210 [0.134772, 0.145649]	100/100
Scaled ILR	89.79 [89.48, 90.09]	0.077231 [0.076164, 0.078298]	100/100
Group ILR	89.85 [89.53, 90.17]	0.077465 [0.076410, 0.078519]	100/100
Coordinate marginal	88.92 [88.45, 89.40]	0.093462 [0.091430, 0.095494]	100/100
Bonferroni	99.55 [99.45, 99.64]	0.297946 [0.286780, 0.309112]	100/100
heterogeneous_k3; minority; K=3, n_cal=499, 1-alpha=0.9			
Euclidean	25.01 [24.18, 25.84]	0.159351 [0.154803, 0.163900]	100/100
Total variation	24.79 [23.96, 25.62]	0.157996 [0.153602, 0.162390]	100/100
Hellinger	18.27 [17.50, 19.03]	0.134628 [0.130042, 0.139214]	100/100
ILR	16.10 [15.34, 16.86]	0.132282 [0.126781, 0.137784]	100/100
Scaled ILR	89.21 [88.42, 89.99]	0.825718 [0.816912, 0.834523]	100/100
Group ILR	90.44 [89.55, 91.34]	0.839674 [0.829639, 0.849709]	100/100
Coordinate marginal	15.05 [14.45, 15.64]	0.092330 [0.090358, 0.094302]	100/100
Bonferroni	44.38 [42.92, 45.84]	0.287207 [0.276574, 0.297840]	100/100
heterogeneous_k10; majority; K=10, n_cal=499, 1-alpha=0.9			
Euclidean	99.42 [99.28, 99.56]	—	100/100
Total variation	99.42 [99.26, 99.57]	—	100/100
Hellinger	99.50 [99.35, 99.65]	—	100/100
ILR	99.48 [99.33, 99.64]	—	100/100
Scaled ILR	89.93 [89.64, 90.22]	—	100/100
Group ILR	89.93 [89.62, 90.23]	—	100/100
Coordinate marginal	61.35 [60.50, 62.19]	—	100/100
Bonferroni	99.78 [99.73, 99.83]	—	100/100
heterogeneous_k10; minority; K=10, n_cal=499, 1-alpha=0.9			
Euclidean	6.22 [4.75, 7.69]	—	100/100
Total variation	5.56 [4.11, 7.01]	—	100/100
Hellinger	5.28 [3.83, 6.73]	—	100/100
ILR	5.11 [3.71, 6.52]	—	100/100
Scaled ILR	88.85 [88.08, 89.61]	—	100/100
Group ILR	90.96 [90.08, 91.84]	—	100/100
Coordinate marginal	0.01 [-0.002, 0.03]	—	100/100
Bonferroni	24.37 [23.24, 25.50]	—	100/100

Table 11 (continued)

Method	Joint coverage (%)	Area fraction	R_c/R
boundary_k3; majority; K=3, n_cal=499, 1-alpha=0.9			
Euclidean	89.69 [89.42, 89.97]	0.074884 [0.072663, 0.077104]	100/100
Total variation	89.71 [89.43, 89.98]	0.073967 [0.071845, 0.076089]	100/100
Hellinger	89.71 [89.42, 90.01]	0.068505 [0.066420, 0.070591]	100/100
ILR	89.91 [89.61, 90.22]	0.127156 [0.122784, 0.131528]	100/100
Coordinate marginal	81.37 [81.03, 81.72]	0.039523 [0.038517, 0.040529]	100/100
Bonferroni	93.46 [93.24, 93.68]	0.118896 [0.115366, 0.122426]	100/100
boundary_k3; minority; K=3, n_cal=499, 1-alpha=0.9			
Euclidean	92.10 [91.63, 92.57]	0.064253 [0.062494, 0.066013]	100/100
Total variation	92.11 [91.66, 92.57]	0.063549 [0.061846, 0.065253]	100/100
Hellinger	91.93 [91.47, 92.39]	0.054224 [0.052649, 0.055798]	100/100
ILR	89.75 [89.22, 90.29]	0.091819 [0.088287, 0.095350]	100/100
Coordinate marginal	85.62 [85.04, 86.21]	0.034262 [0.033468, 0.035055]	100/100
Bonferroni	95.12 [94.74, 95.51]	0.107799 [0.104717, 0.110882]	100/100
boundary_k10; majority; K=10, n_cal=499, 1-alpha=0.9			
Euclidean	89.90 [89.59, 90.22]	—	100/100
Total variation	89.84 [89.52, 90.16]	—	100/100
Hellinger	89.92 [89.61, 90.23]	—	100/100
ILR	89.94 [89.71, 90.18]	—	100/100
Coordinate marginal	45.97 [45.58, 46.36]	—	100/100
Bonferroni	91.08 [90.79, 91.36]	—	100/100
boundary_k10; minority; K=10, n_cal=499, 1-alpha=0.9			
Euclidean	91.02 [90.50, 91.54]	—	100/100
Total variation	91.40 [90.92, 91.88]	—	100/100
Hellinger	91.14 [90.68, 91.60]	—	100/100
ILR	90.04 [89.58, 90.50]	—	100/100
Coordinate marginal	51.87 [51.14, 52.60]	—	100/100
Bonferroni	92.38 [91.89, 92.87]	—	100/100
dirichlet_k3; majority; K=3, n_cal=499, 1-alpha=0.9			
Euclidean	89.77 [89.47, 90.07]	0.113867 [0.112123, 0.115611]	100/100
Total variation	89.86 [89.56, 90.15]	0.113673 [0.111999, 0.115347]	100/100
Hellinger	89.84 [89.55, 90.14]	0.115390 [0.113402, 0.117378]	100/100
ILR	90.17 [89.91, 90.42]	0.292784 [0.286085, 0.299483]	100/100
Coordinate marginal	79.43 [79.04, 79.83]	0.078588 [0.077346, 0.079830]	100/100
Bonferroni	92.71 [92.45, 92.96]	0.135134 [0.133067, 0.137201]	100/100

Table 11 (continued)

Method	Joint coverage (%)	Area fraction	R_c/R
dirichlet_k3; minority; K=3, n_cal=499, 1-alpha=0.9			
Euclidean	90.67 [90.15, 91.19]	0.105480 [0.103950, 0.107011]	100/100
Total variation	90.86 [90.32, 91.39]	0.105551 [0.104062, 0.107041]	100/100
Hellinger	89.81 [89.28, 90.35]	0.105467 [0.103681, 0.107253]	100/100
ILR	88.96 [88.42, 89.50]	0.266017 [0.260097, 0.271938]	100/100
Coordinate marginal	80.93 [80.23, 81.62]	0.073339 [0.072278, 0.074400]	100/100
Bonferroni	93.41 [93.01, 93.82]	0.126853 [0.124998, 0.128709]	100/100
dirichlet_k10; majority; K=10, n_cal=499, 1-alpha=0.9			
Euclidean	89.87 [89.54, 90.20]	—	100/100
Total variation	89.69 [89.35, 90.04]	—	100/100
Hellinger	89.97 [89.66, 90.28]	—	100/100
ILR	90.30 [90.02, 90.57]	—	100/100
Coordinate marginal	33.91 [33.48, 34.33]	—	100/100
Bonferroni	90.67 [90.36, 90.98]	—	100/100
dirichlet_k10; minority; K=10, n_cal=499, 1-alpha=0.9			
Euclidean	90.40 [89.80, 90.99]	—	100/100
Total variation	90.80 [90.25, 91.34]	—	100/100
Hellinger	89.95 [89.40, 90.51]	—	100/100
ILR	87.71 [87.14, 88.29]	—	100/100
Coordinate marginal	36.64 [35.75, 37.53]	—	100/100
Bonferroni	91.55 [91.07, 92.02]	—	100/100
zeros_k3; majority; K=3, n_cal=499, 1-alpha=0.9, epsilon=0.001			
Euclidean	90.16 [89.87, 90.46]	0.759906 [0.752972, 0.766839]	100/100
Total variation	90.16 [89.87, 90.46]	0.762759 [0.756130, 0.769389]	100/100
Hellinger	90.02 [89.74, 90.30]	0.892447 [0.887292, 0.897602]	100/100
Smoothed ILR	89.88 [89.62, 90.15]	0.997200 [0.997028, 0.997373]	100/100
Coordinate marginal	78.34 [77.96, 78.73]	0.569292 [0.563004, 0.575580]	100/100
Bonferroni	92.54 [92.30, 92.78]	0.822996 [0.816997, 0.828995]	100/100
ILR (undefined)	Undefined at exact zeros		
zeros_k3; minority; K=3, n_cal=499, 1-alpha=0.9, epsilon=0.001			
Euclidean	90.25 [89.71, 90.79]	0.733007 [0.725359, 0.740655]	100/100
Total variation	90.32 [89.82, 90.82]	0.736002 [0.728798, 0.743205]	100/100
Hellinger	90.52 [90.05, 91.00]	0.872194 [0.866287, 0.878102]	100/100
Smoothed ILR	90.53 [90.04, 91.03]	0.996453 [0.996232, 0.996675]	100/100
Coordinate marginal	79.22 [78.58, 79.86]	0.549101 [0.542272, 0.555930]	100/100
Bonferroni	92.74 [92.32, 93.16]	0.796877 [0.790208, 0.803546]	100/100
ILR (undefined)	Undefined at exact zeros		

Table 11 (continued)

Method	Joint coverage (%)	Area fraction	R_c/R
zeros_k3; zero_face; K=3, n_cal=499, 1-alpha=0.9, epsilon=0.001			
Euclidean	69.26 [68.54, 69.98]	—	100/100
Total variation	69.38 [68.66, 70.10]	—	100/100
Hellinger	61.15 [60.14, 62.17]	—	100/100
Smoothed ILR	59.85 [58.78, 60.92]	—	100/100
Coordinate marginal	47.89 [47.17, 48.61]	—	100/100
Bonferroni	75.62 [74.95, 76.29]	—	100/100
ILR (undefined)	Undefined at exact zeros		
zeros_k3; positive; K=3, n_cal=499, 1-alpha=0.9, epsilon=0.001			
Euclidean	97.15 [96.97, 97.33]	—	100/100
Total variation	97.12 [96.95, 97.30]	—	100/100
Hellinger	99.73 [99.69, 99.77]	—	100/100
Smoothed ILR	100.00 [100.00, 100.00]	—	100/100
Coordinate marginal	88.64 [88.30, 88.98]	—	100/100
Bonferroni	98.21 [98.08, 98.34]	—	100/100
ILR (undefined)	Undefined at exact zeros		
zeros_k10; majority; K=10, n_cal=499, 1-alpha=0.9, epsilon=0.001			
Euclidean	89.95 [89.67, 90.22]	—	100/100
Total variation	89.95 [89.67, 90.22]	—	100/100
Hellinger	90.01 [89.71, 90.30]	—	100/100
Smoothed ILR	89.93 [89.63, 90.23]	—	100/100
Coordinate marginal	26.74 [26.39, 27.10]	—	100/100
Bonferroni	89.95 [89.63, 90.26]	—	100/100
ILR (undefined)	Undefined at exact zeros		
zeros_k10; minority; K=10, n_cal=499, 1-alpha=0.9, epsilon=0.001			
Euclidean	89.02 [88.51, 89.54]	—	100/100
Total variation	89.18 [88.66, 89.70]	—	100/100
Hellinger	89.63 [89.13, 90.12]	—	100/100
Smoothed ILR	90.09 [89.55, 90.63]	—	100/100
Coordinate marginal	26.27 [25.58, 26.95]	—	100/100
Bonferroni	89.49 [89.02, 89.95]	—	100/100
ILR (undefined)	Undefined at exact zeros		
zeros_k10; zero_face; K=10, n_cal=499, 1-alpha=0.9, epsilon=0.001			
Euclidean	85.29 [84.83, 85.76]	—	100/100
Total variation	83.67 [83.19, 84.15]	—	100/100
Hellinger	72.25 [71.51, 72.99]	—	100/100
Smoothed ILR	59.82 [58.66, 60.99]	—	100/100
Coordinate marginal	17.37 [16.89, 17.84]	—	100/100
Bonferroni	87.20 [86.77, 87.64]	—	100/100
ILR (undefined)	Undefined at exact zeros		
zeros_k10; positive; K=10, n_cal=499, 1-alpha=0.9, epsilon=0.001			
Euclidean	91.38 [91.13, 91.63]	—	100/100
Total variation	91.94 [91.69, 92.18]	—	100/100
Hellinger	95.89 [95.72, 96.05]	—	100/100
Smoothed ILR	100.00 [100.00, 100.00]	—	100/100
Coordinate marginal	29.80 [29.45, 30.16]	—	100/100
Bonferroni	90.80 [90.50, 91.10]	—	100/100
ILR (undefined)	Undefined at exact zeros		

Table 11 (continued)

Method	Joint coverage (%)	Area fraction	R_c/R
interior_ncal99; majority; K=3, n_cal=99, 1-alpha=0.9			
Euclidean	89.61 [88.97, 90.26]	0.274910 [0.267451, 0.282368]	100/100
Total variation	89.36 [88.74, 89.99]	0.273142 [0.265968, 0.280315]	100/100
Hellinger	89.74 [89.13, 90.35]	0.233218 [0.227159, 0.239277]	100/100
ILR	89.39 [88.77, 90.00]	0.227242 [0.221574, 0.232910]	100/100
Coordinate marginal	77.46 [76.73, 78.20]	0.178571 [0.174474, 0.182668]	100/100
Bonferroni	92.06 [91.53, 92.59]	0.309574 [0.301784, 0.317363]	100/100
interior_ncal99; minority; K=3, n_cal=99, 1-alpha=0.9			
Euclidean	90.40 [89.72, 91.09]	0.263506 [0.256141, 0.270870]	100/100
Total variation	90.22 [89.54, 90.91]	0.262150 [0.255036, 0.269263]	100/100
Hellinger	90.34 [89.66, 91.02]	0.223250 [0.217244, 0.229256]	100/100
ILR	89.40 [88.64, 90.16]	0.214371 [0.208796, 0.219946]	100/100
Coordinate marginal	79.15 [78.21, 80.10]	0.172253 [0.168439, 0.176067]	100/100
Bonferroni	92.74 [92.14, 93.34]	0.297155 [0.289734, 0.304575]	100/100
interior_ncal1999; majority; K=3, n_cal=1999, 1-alpha=0.9			
Euclidean	89.96 [89.75, 90.17]	0.273040 [0.270994, 0.275085]	100/100
Total variation	89.83 [89.62, 90.05]	0.273551 [0.271662, 0.275440]	100/100
Hellinger	89.94 [89.73, 90.16]	0.230833 [0.228917, 0.232748]	100/100
ILR	89.88 [89.67, 90.09]	0.228932 [0.226676, 0.231188]	100/100
Coordinate marginal	78.08 [77.80, 78.36]	0.179239 [0.178074, 0.180403]	100/100
Bonferroni	91.92 [91.71, 92.13]	0.297694 [0.295467, 0.299921]	100/100
interior_ncal1999; minority; K=3, n_cal=1999, 1-alpha=0.9			
Euclidean	90.53 [90.09, 90.97]	0.261560 [0.259300, 0.263820]	100/100
Total variation	90.52 [90.03, 91.00]	0.262481 [0.260260, 0.264702]	100/100
Hellinger	90.42 [89.98, 90.86]	0.220903 [0.218797, 0.223009]	100/100
ILR	90.05 [89.60, 90.50]	0.215961 [0.213283, 0.218640]	100/100
Coordinate marginal	79.80 [79.15, 80.46]	0.172704 [0.171498, 0.173911]	100/100
Bonferroni	92.55 [92.14, 92.95]	0.286468 [0.283820, 0.289115]	100/100

Table 11 (continued)

Method	Joint coverage (%)	Area fraction	R_c/R
interior_alpha05; majority; K=3, n_cal=499, 1-alpha=0.95			
Euclidean	94.85 [94.60, 95.10]	0.349323 [0.344473, 0.354174]	100/100
Total variation	94.84 [94.59, 95.08]	0.348873 [0.344193, 0.353553]	100/100
Hellinger	94.85 [94.60, 95.10]	0.295888 [0.291732, 0.300044]	100/100
ILR	94.97 [94.74, 95.20]	0.287235 [0.283415, 0.291055]	100/100
Coordinate marginal	88.07 [87.76, 88.38]	0.253882 [0.251060, 0.256704]	100/100
Bonferroni	95.84 [95.64, 96.03]	0.370903 [0.366020, 0.375786]	100/100
interior_alpha05; minority; K=3, n_cal=499, 1-alpha=0.95			
Euclidean	95.17 [94.82, 95.51]	0.335184 [0.329847, 0.340520]	100/100
Total variation	95.23 [94.87, 95.59]	0.335496 [0.330349, 0.340643]	100/100
Hellinger	95.23 [94.89, 95.57]	0.283327 [0.278725, 0.287929]	100/100
ILR	95.01 [94.65, 95.36]	0.271787 [0.267294, 0.276281]	100/100
Coordinate marginal	89.15 [88.62, 89.68]	0.244463 [0.241219, 0.247707]	100/100
Bonferroni	96.28 [95.98, 96.59]	0.356731 [0.351231, 0.362232]	100/100
heterogeneous_ncal99; majority; K=3, n_cal=99, 1-alpha=0.9			
Euclidean	97.21 [96.74, 97.67]	0.188111 [0.175831, 0.200391]	100/100
Total variation	97.30 [96.86, 97.73]	0.190808 [0.178319, 0.203298]	100/100
Hellinger	97.81 [97.35, 98.27]	0.179013 [0.162466, 0.195560]	100/100
ILR	98.08 [97.66, 98.51]	0.190727 [0.169008, 0.212446]	100/100
Scaled ILR	89.68 [89.04, 90.32]	0.078337 [0.076180, 0.080495]	100/100
Group ILR	90.07 [89.48, 90.66]	0.079542 [0.077472, 0.081612]	100/100
Coordinate marginal	89.42 [88.49, 90.36]	0.103165 [0.097651, 0.108678]	100/100
Bonferroni	99.31 [99.09, 99.53]	0.359380 [0.334889, 0.383872]	100/100
heterogeneous_ncal99; minority; K=3, n_cal=99, 1-alpha=0.9			
Euclidean	28.44 [26.59, 30.28]	0.181585 [0.169812, 0.193358]	100/100
Total variation	28.56 [26.72, 30.39]	0.184064 [0.172052, 0.196076]	100/100
Hellinger†	23.49 [21.26, 25.72]	0.172060 [0.155715, 0.188406]	100/100
ILR†	21.34 [18.93, 23.75]	0.180799 [0.159709, 0.201889]	100/100
Scaled ILR	89.39 [88.56, 90.22]	0.827176 [0.817875, 0.836478]	100/100
Group ILR	94.93 [93.64, 96.22]	0.910520 [0.892566, 0.928473]	100/100
Coordinate marginal	16.69 [15.62, 17.75]	0.101410 [0.096151, 0.106668]	100/100
Bonferroni†	49.99 [47.17, 52.80]	0.346383 [0.322866, 0.369899]	100/100

Table 11 (continued)

Method	Joint coverage (%)	Area fraction	R_c/R
heterogeneous_ncal1999; majority; K=3, n_cal=1999, 1-alpha=0.9			
Euclidean	97.12 [96.97, 97.26]	0.161695 [0.159649, 0.163742]	100/100
Total variation	97.17 [97.03, 97.31]	0.162903 [0.160885, 0.164921]	100/100
Hellinger	97.94 [97.81, 98.08]	0.138614 [0.136481, 0.140746]	100/100
ILR	98.26 [98.13, 98.39]	0.134467 [0.132068, 0.136865]	100/100
Scaled ILR	89.82 [89.61, 90.03]	0.077064 [0.076332, 0.077796]	100/100
Group ILR	89.82 [89.62, 90.03]	0.077092 [0.076385, 0.077799]	100/100
Coordinate marginal	89.41 [89.15, 89.67]	0.094201 [0.093226, 0.095176]	100/100
Bonferroni	99.61 [99.56, 99.66]	0.281494 [0.276415, 0.286573]	100/100
heterogeneous_ncal1999; minority; K=3, n_cal=1999, 1-alpha=0.9			
Euclidean	24.87 [24.22, 25.51]	0.156338 [0.154066, 0.158611]	100/100
Total variation	24.76 [24.13, 25.39]	0.157316 [0.155039, 0.159593]	100/100
Hellinger	18.09 [17.57, 18.62]	0.132931 [0.130387, 0.135474]	100/100
ILR	15.50 [15.01, 15.99]	0.126797 [0.123786, 0.129808]	100/100
Scaled ILR	89.27 [88.49, 90.05]	0.825646 [0.817233, 0.834059]	100/100
Group ILR	89.43 [88.82, 90.03]	0.827347 [0.821740, 0.832953]	100/100
Coordinate marginal	15.19 [14.68, 15.70]	0.093068 [0.092053, 0.094083]	100/100
Bonferroni	42.98 [42.13, 43.83]	0.271792 [0.266099, 0.277485]	100/100
heterogeneous_alpha05; majority; K=3, n_cal=499, 1-alpha=0.95			
Euclidean	99.90 [99.86, 99.93]	0.364092 [0.348489, 0.379695]	100/100
Total variation	99.90 [99.87, 99.93]	0.364638 [0.349250, 0.380027]	100/100
Hellinger	99.98 [99.97, 99.99]	0.385621 [0.369639, 0.401604]	100/100
ILR	100.00 [100.00, 100.00]	0.448406 [0.431470, 0.465341]	100/100
Scaled ILR	94.79 [94.56, 95.01]	0.099124 [0.097572, 0.100677]	100/100
Group ILR	94.90 [94.67, 95.13]	0.099849 [0.098328, 0.101371]	100/100
Coordinate marginal	97.95 [97.70, 98.19]	0.192232 [0.185849, 0.198616]	100/100
Bonferroni	99.97 [99.96, 99.99]	0.514038 [0.497844, 0.530233]	100/100

Table 11 (continued)

Method	Joint coverage (%)	Area fraction	R_c/R
heterogeneous_alpha05; minority; K=3, n_cal=499, 1-alpha=0.95			
Euclidean	51.26 [49.39, 53.13]	0.349157 [0.334267, 0.364047]	100/100
Total variation	50.89 [49.08, 52.70]	0.350407 [0.335568, 0.365246]	100/100
Hellinger†	50.15 [48.13, 52.16]	0.369546 [0.354315, 0.384777]	100/100
ILR	49.32 [47.41, 51.22]	0.428858 [0.411911, 0.445805]	100/100
Scaled ILR	94.46 [93.94, 94.98]	0.887288 [0.880382, 0.894194]	100/100
Group ILR	95.53 [94.90, 96.16]	0.903454 [0.895496, 0.911413]	100/100
Coordinate marginal	30.54 [29.50, 31.58]	0.185820 [0.179748, 0.191893]	100/100
Bonferroni	66.41 [64.85, 67.97]	0.495797 [0.479965, 0.511630]	100/100
heterogeneous_shift; majority; K=3, n_cal=499, 1-alpha=0.9, group-prevalence shift 0.10 to 0.40			
Euclidean	97.09 [96.81, 97.37]	0.164711 [0.160126, 0.169295]	100/100
Total variation	96.98 [96.69, 97.26]	0.163415 [0.158975, 0.167856]	100/100
Hellinger	97.79 [97.53, 98.05]	0.140085 [0.135771, 0.144399]	100/100
ILR	98.23 [98.00, 98.46]	0.139856 [0.134454, 0.145257]	100/100
Scaled ILR	89.84 [89.54, 90.15]	0.077031 [0.075939, 0.078123]	100/100
Group ILR	89.90 [89.58, 90.22]	0.077269 [0.076167, 0.078371]	100/100
Coordinate marginal	89.03 [88.56, 89.49]	0.093437 [0.091412, 0.095463]	100/100
Bonferroni	99.56 [99.46, 99.66]	0.297693 [0.286574, 0.308811]	100/100
heterogeneous_shift; minority; K=3, n_cal=499, 1-alpha=0.9, group-prevalence shift 0.10 to 0.40			
Euclidean	25.14 [24.37, 25.90]	0.158932 [0.154469, 0.163394]	100/100
Total variation	24.85 [24.11, 25.58]	0.157568 [0.153275, 0.161861]	100/100
Hellinger	18.45 [17.78, 19.12]	0.133865 [0.129509, 0.138221]	100/100
ILR	15.92 [15.25, 16.60]	0.131151 [0.125841, 0.136460]	100/100
Scaled ILR	89.57 [88.82, 90.31]	0.824741 [0.816225, 0.833258]	100/100
Group ILR	90.65 [89.78, 91.52]	0.838566 [0.828626, 0.848506]	100/100
Coordinate marginal	15.12 [14.68, 15.55]	0.092287 [0.090325, 0.094249]	100/100
Bonferroni	44.46 [43.05, 45.86]	0.286531 [0.275584, 0.297478]	100/100
zeros_eps1e-06; majority; K=3, n_cal=499, 1-alpha=0.9, epsilon=1e-06			
Euclidean	90.16 [89.87, 90.46]	0.759906 [0.752972, 0.766839]	100/100
Total variation	90.16 [89.87, 90.46]	0.762759 [0.756130, 0.769389]	100/100
Hellinger	90.02 [89.74, 90.30]	0.892447 [0.887292, 0.897602]	100/100
Smoothed ILR	89.88 [89.61, 90.16]	1.000021 [0.999995, 1.000046]	100/100
Coordinate marginal	78.34 [77.96, 78.73]	0.569292 [0.563004, 0.575580]	100/100
Bonferroni	92.54 [92.30, 92.78]	0.822996 [0.816997, 0.828995]	100/100
ILR (undefined)	Undefined at exact zeros		

Table 11 (continued)

Method	Joint coverage (%)	Area fraction	R_c/R
zeros_eps1e-06; minority; K=3, n_cal=499, 1-alpha=0.9, epsilon=1e-06			
Euclidean	90.25 [89.71, 90.79]	0.733007 [0.725359, 0.740655]	100/100
Total variation	90.32 [89.82, 90.82]	0.736002 [0.728798, 0.743205]	100/100
Hellinger	90.52 [90.05, 91.00]	0.872194 [0.866287, 0.878102]	100/100
Smoothed ILR	90.48 [89.98, 90.99]	0.999991 [0.999959, 1.000024]	100/100
Coordinate marginal	79.22 [78.58, 79.86]	0.549101 [0.542272, 0.555930]	100/100
Bonferroni	92.74 [92.32, 93.16]	0.796877 [0.790208, 0.803546]	100/100
ILR (undefined)	Undefined at exact zeros		
zeros_eps1e-06; zero_face; K=3, n_cal=499, 1-alpha=0.9, epsilon=1e-06			
Euclidean	69.26 [68.54, 69.98]	—	100/100
Total variation	69.38 [68.66, 70.10]	—	100/100
Hellinger	61.15 [60.14, 62.17]	—	100/100
Smoothed ILR	59.84 [58.75, 60.93]	—	100/100
Coordinate marginal	47.89 [47.17, 48.61]	—	100/100
Bonferroni	75.62 [74.95, 76.29]	—	100/100
ILR (undefined)	Undefined at exact zeros		
zeros_eps1e-06; positive; K=3, n_cal=499, 1-alpha=0.9, epsilon=1e-06			
Euclidean	97.15 [96.97, 97.33]	—	100/100
Total variation	97.12 [96.95, 97.30]	—	100/100
Hellinger	99.73 [99.69, 99.77]	—	100/100
Smoothed ILR	100.00 [100.00, 100.00]	—	100/100
Coordinate marginal	88.64 [88.30, 88.98]	—	100/100
Bonferroni	98.21 [98.08, 98.34]	—	100/100
ILR (undefined)	Undefined at exact zeros		
zeros_eps0.01; majority; K=3, n_cal=499, 1-alpha=0.9, epsilon=0.01			
Euclidean	90.16 [89.87, 90.46]	0.759906 [0.752972, 0.766839]	100/100
Total variation	90.16 [89.87, 90.46]	0.762759 [0.756130, 0.769389]	100/100
Hellinger	90.02 [89.74, 90.30]	0.892447 [0.887292, 0.897602]	100/100
Smoothed ILR	89.93 [89.66, 90.21]	0.969610 [0.967905, 0.971315]	100/100
Coordinate marginal	78.34 [77.96, 78.73]	0.569292 [0.563004, 0.575580]	100/100
Bonferroni	92.54 [92.30, 92.78]	0.822996 [0.816997, 0.828995]	100/100
ILR (undefined)	Undefined at exact zeros		
zeros_eps0.01; minority; K=3, n_cal=499, 1-alpha=0.9, epsilon=0.01			
Euclidean	90.25 [89.71, 90.79]	0.733007 [0.725359, 0.740655]	100/100
Total variation	90.32 [89.82, 90.82]	0.736002 [0.728798, 0.743205]	100/100
Hellinger	90.52 [90.05, 91.00]	0.872194 [0.866287, 0.878102]	100/100
Smoothed ILR	90.52 [90.04, 90.99]	0.962773 [0.960664, 0.964883]	100/100
Coordinate marginal	79.22 [78.58, 79.86]	0.549101 [0.542272, 0.555930]	100/100
Bonferroni	92.74 [92.32, 93.16]	0.796877 [0.790208, 0.803546]	100/100
ILR (undefined)	Undefined at exact zeros		

Table 11 (continued)

Method	Joint coverage (%)	Area fraction	R_c/R
zeros_eps0.01; zero_face; K=3, n_cal=499, 1-alpha=0.9, epsilon=0.01			
Euclidean	69.26 [68.54, 69.98]	—	100/100
Total variation	69.38 [68.66, 70.10]	—	100/100
Hellinger	61.15 [60.14, 62.17]	—	100/100
Smoothed ILR	60.03 [58.94, 61.12]	—	100/100
Coordinate marginal	47.89 [47.17, 48.61]	—	100/100
Bonferroni	75.62 [74.95, 76.29]	—	100/100
ILR (undefined)	Undefined at exact zeros		
zeros_eps0.01; positive; K=3, n_cal=499, 1-alpha=0.9, epsilon=0.01			
Euclidean	97.15 [96.97, 97.33]	—	100/100
Total variation	97.12 [96.95, 97.30]	—	100/100
Hellinger	99.73 [99.69, 99.77]	—	100/100
Smoothed ILR	100.00 [100.00, 100.00]	—	100/100
Coordinate marginal	88.64 [88.30, 88.98]	—	100/100
Bonferroni	98.21 [98.08, 98.34]	—	100/100
ILR (undefined)	Undefined at exact zeros		

Table 12: Full-domain and infinite-threshold audit (percent), averaged across independent trials and test observations within each trial. Full-domain entries are a certified percentage or [lower, upper] percentages when smoothed-ILR status is unresolved; this bracket is not a confidence interval. Unknown is the unresolved-status percentage. An infinite calibration threshold suffices for a full-domain region, but finite thresholds can also cover the domain. Outer-trial confidence intervals for each rate and certified bound are retained in the exact-value companion CSV; no rate uncertainty is implied by these brackets.

Method	Full domain (%)	Unknown (%)	Infinite (%, 95% CI)
interior_k3; all; K=3, n_cal=499, 1-alpha=0.9			
Euclidean	0.00	0.00	0.00 [0.00, 0.00]
Total variation	0.00	0.00	0.00 [0.00, 0.00]
Hellinger	0.00	0.00	0.00 [0.00, 0.00]
ILR	0.00	0.00	0.00 [0.00, 0.00]
Coordinate marginal	0.00	0.00	0.00 [0.00, 0.00]
Bonferroni	0.00	0.00	0.00 [0.00, 0.00]
interior_k10; all; K=10, n_cal=499, 1-alpha=0.9			
Euclidean	0.00	0.00	0.00 [0.00, 0.00]
Total variation	0.00	0.00	0.00 [0.00, 0.00]
Hellinger	0.00	0.00	0.00 [0.00, 0.00]
ILR	0.00	0.00	0.00 [0.00, 0.00]
Coordinate marginal	0.00	0.00	0.00 [0.00, 0.00]
Bonferroni	0.00	0.00	0.00 [0.00, 0.00]
heterogeneous_k3; all; K=3, n_cal=499, 1-alpha=0.9			
Euclidean	0.00	0.00	0.00 [0.00, 0.00]
Total variation	0.00	0.00	0.00 [0.00, 0.00]
Hellinger	0.00	0.00	0.00 [0.00, 0.00]
ILR	0.00	0.00	0.00 [0.00, 0.00]
Scaled ILR	0.00	0.00	0.00 [0.00, 0.00]
Group ILR	0.00	0.00	0.00 [0.00, 0.00]
Coordinate marginal	0.00	0.00	0.00 [0.00, 0.00]
Bonferroni	0.00	0.00	0.00 [0.00, 0.00]
heterogeneous_k10; all; K=10, n_cal=499, 1-alpha=0.9			
Euclidean	0.00	0.00	0.00 [0.00, 0.00]
Total variation	0.00	0.00	0.00 [0.00, 0.00]
Hellinger	0.00	0.00	0.00 [0.00, 0.00]
ILR	0.00	0.00	0.00 [0.00, 0.00]
Scaled ILR	0.00	0.00	0.00 [0.00, 0.00]
Group ILR	0.00	0.00	0.00 [0.00, 0.00]
Coordinate marginal	0.00	0.00	0.00 [0.00, 0.00]
Bonferroni	0.00	0.00	0.00 [0.00, 0.00]
boundary_k3; all; K=3, n_cal=499, 1-alpha=0.9			
Euclidean	0.00	0.00	0.00 [0.00, 0.00]
Total variation	0.00	0.00	0.00 [0.00, 0.00]
Hellinger	0.00	0.00	0.00 [0.00, 0.00]
ILR	0.00	0.00	0.00 [0.00, 0.00]
Coordinate marginal	0.00	0.00	0.00 [0.00, 0.00]
Bonferroni	0.00	0.00	0.00 [0.00, 0.00]
boundary_k10; all; K=10, n_cal=499, 1-alpha=0.9			
Euclidean	0.00	0.00	0.00 [0.00, 0.00]
Total variation	0.00	0.00	0.00 [0.00, 0.00]
Hellinger	0.00	0.00	0.00 [0.00, 0.00]
ILR	0.00	0.00	0.00 [0.00, 0.00]
Coordinate marginal	0.00	0.00	0.00 [0.00, 0.00]
Bonferroni	0.00	0.00	0.00 [0.00, 0.00]

Table 12 (continued)

Method	Full domain (%)	Unknown (%)	Infinite (%, 95% CI)
dirichlet_k3; all; K=3, n_cal=499, 1-alpha=0.9			
Euclidean	0.00	0.00	0.00 [0.00, 0.00]
Total variation	0.00	0.00	0.00 [0.00, 0.00]
Hellinger	0.00	0.00	0.00 [0.00, 0.00]
ILR	0.00	0.00	0.00 [0.00, 0.00]
Coordinate marginal	0.00	0.00	0.00 [0.00, 0.00]
Bonferroni	0.00	0.00	0.00 [0.00, 0.00]
dirichlet_k10; all; K=10, n_cal=499, 1-alpha=0.9			
Euclidean	0.00	0.00	0.00 [0.00, 0.00]
Total variation	0.00	0.00	0.00 [0.00, 0.00]
Hellinger	0.00	0.00	0.00 [0.00, 0.00]
ILR	0.00	0.00	0.00 [0.00, 0.00]
Coordinate marginal	0.00	0.00	0.00 [0.00, 0.00]
Bonferroni	0.00	0.00	0.00 [0.00, 0.00]
zeros_k3; all; K=3, n_cal=499, 1-alpha=0.9, epsilon=0.001			
Euclidean	0.00	0.00	0.00 [0.00, 0.00]
Total variation	0.00	0.00	0.00 [0.00, 0.00]
Hellinger	0.00	0.00	0.00 [0.00, 0.00]
Smoothed ILR	0.00	0.00	0.00 [0.00, 0.00]
Coordinate marginal	0.00	0.00	0.00 [0.00, 0.00]
Bonferroni	0.00	0.00	0.00 [0.00, 0.00]
ILR (undefined)	Undefined at exact zeros		
zeros_k10; all; K=10, n_cal=499, 1-alpha=0.9, epsilon=0.001			
Euclidean	0.00	0.00	0.00 [0.00, 0.00]
Total variation	0.00	0.00	0.00 [0.00, 0.00]
Hellinger	0.00	0.00	0.00 [0.00, 0.00]
Smoothed ILR	0.00	0.00	0.00 [0.00, 0.00]
Coordinate marginal	0.00	0.00	0.00 [0.00, 0.00]
Bonferroni	0.00	0.00	0.00 [0.00, 0.00]
ILR (undefined)	Undefined at exact zeros		
interior_ncal99; all; K=3, n_cal=99, 1-alpha=0.9			
Euclidean	0.00	0.00	0.00 [0.00, 0.00]
Total variation	0.00	0.00	0.00 [0.00, 0.00]
Hellinger	0.00	0.00	0.00 [0.00, 0.00]
ILR	0.00	0.00	0.00 [0.00, 0.00]
Coordinate marginal	0.00	0.00	0.00 [0.00, 0.00]
Bonferroni	0.00	0.00	0.00 [0.00, 0.00]
interior_ncal1999; all; K=3, n_cal=1999, 1-alpha=0.9			
Euclidean	0.00	0.00	0.00 [0.00, 0.00]
Total variation	0.00	0.00	0.00 [0.00, 0.00]
Hellinger	0.00	0.00	0.00 [0.00, 0.00]
ILR	0.00	0.00	0.00 [0.00, 0.00]
Coordinate marginal	0.00	0.00	0.00 [0.00, 0.00]
Bonferroni	0.00	0.00	0.00 [0.00, 0.00]
interior_alpha05; all; K=3, n_cal=499, 1-alpha=0.95			
Euclidean	0.00	0.00	0.00 [0.00, 0.00]
Total variation	0.00	0.00	0.00 [0.00, 0.00]
Hellinger	0.00	0.00	0.00 [0.00, 0.00]
ILR	0.00	0.00	0.00 [0.00, 0.00]
Coordinate marginal	0.00	0.00	0.00 [0.00, 0.00]
Bonferroni	0.00	0.00	0.00 [0.00, 0.00]

Table 12 (continued)

Method	Full domain (%)	Unknown (%)	Infinite (%, 95% CI)
heterogeneous_ncal99; all; K=3, n_cal=99, 1-alpha=0.9			
Euclidean	0.00	0.00	0.00 [0.00, 0.00]
Total variation	0.00	0.00	0.00 [0.00, 0.00]
Hellinger	0.00	0.00	0.00 [0.00, 0.00]
ILR	0.00	0.00	0.00 [0.00, 0.00]
Scaled ILR	0.00	0.00	0.00 [0.00, 0.00]
Group ILR	3.03	0.00	3.03 [2.13, 3.93]
Coordinate marginal	0.00	0.00	0.00 [0.00, 0.00]
Bonferroni	0.00	0.00	0.00 [0.00, 0.00]
heterogeneous_ncal1999; all; K=3, n_cal=1999, 1-alpha=0.9			
Euclidean	0.00	0.00	0.00 [0.00, 0.00]
Total variation	0.00	0.00	0.00 [0.00, 0.00]
Hellinger	0.00	0.00	0.00 [0.00, 0.00]
ILR	0.00	0.00	0.00 [0.00, 0.00]
Scaled ILR	0.00	0.00	0.00 [0.00, 0.00]
Group ILR	0.00	0.00	0.00 [0.00, 0.00]
Coordinate marginal	0.00	0.00	0.00 [0.00, 0.00]
Bonferroni	0.00	0.00	0.00 [0.00, 0.00]
heterogeneous_alpha05; all; K=3, n_cal=499, 1-alpha=0.95			
Euclidean	0.00	0.00	0.00 [0.00, 0.00]
Total variation	0.00	0.00	0.00 [0.00, 0.00]
Hellinger	0.00	0.00	0.00 [0.00, 0.00]
ILR	0.00	0.00	0.00 [0.00, 0.00]
Scaled ILR	0.00	0.00	0.00 [0.00, 0.00]
Group ILR	0.00	0.00	0.00 [0.00, 0.00]
Coordinate marginal	0.00	0.00	0.00 [0.00, 0.00]
Bonferroni	0.00	0.00	0.00 [0.00, 0.00]
heterogeneous_shift; all; K=3, n_cal=499, 1-alpha=0.9, group-prevalence shift 0.10 to 0.40			
Euclidean	0.00	0.00	0.00 [0.00, 0.00]
Total variation	0.00	0.00	0.00 [0.00, 0.00]
Hellinger	0.00	0.00	0.00 [0.00, 0.00]
ILR	0.00	0.00	0.00 [0.00, 0.00]
Scaled ILR	0.00	0.00	0.00 [0.00, 0.00]
Group ILR	0.00	0.00	0.00 [0.00, 0.00]
Coordinate marginal	0.00	0.00	0.00 [0.00, 0.00]
Bonferroni	0.00	0.00	0.00 [0.00, 0.00]
zeros_eps1e-06; all; K=3, n_cal=499, 1-alpha=0.9, epsilon=1e-06			
Euclidean	0.00	0.00	0.00 [0.00, 0.00]
Total variation	0.00	0.00	0.00 [0.00, 0.00]
Hellinger	0.00	0.00	0.00 [0.00, 0.00]
Smoothed ILR	0.00	0.00	0.00 [0.00, 0.00]
Coordinate marginal	0.00	0.00	0.00 [0.00, 0.00]
Bonferroni	0.00	0.00	0.00 [0.00, 0.00]
ILR (undefined)	Undefined at exact zeros		
zeros_eps0.01; all; K=3, n_cal=499, 1-alpha=0.9, epsilon=0.01			
Euclidean	0.00	0.00	0.00 [0.00, 0.00]
Total variation	0.00	0.00	0.00 [0.00, 0.00]
Hellinger	0.00	0.00	0.00 [0.00, 0.00]
Smoothed ILR	0.00	0.00	0.00 [0.00, 0.00]
Coordinate marginal	0.00	0.00	0.00 [0.00, 0.00]
Bonferroni	0.00	0.00	0.00 [0.00, 0.00]
ILR (undefined)	Undefined at exact zeros		

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